

Name: _____

Double Award Applied Science – Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision. **You will also need the resource pack paper from the correct year.**

Year	Score /75	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries- Raw Score

Year	G	F	E	D	C	Max UMS
2017	10	14	18	22	27	32
2018	17	21	25	29	34	39
2019	13	17	21	25	30	35
2022	13	17	21	25	30	35
2023	12	17	22	27	33	40

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – NEW

3445U20-1



APPLIED SCIENCE (Double Award)
Unit 2: Space, Health and Life
FOUNDATION TIER

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 30 minutes

Section A	For Examiner's use only			
	Question	Maximum Mark	Mark Awarded	
	1.	9		
	2.	7		
	3.	7		
	4.	15		
	5.	12		
	Section B	6.	6	
		7.	19	
		Total	75	

3445U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 5(a) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

Section A

*Answer **all** questions in the spaces provided.*

1. There is a minimum charge of 5p for carrier bags in Wales. The aim of the charge is to reduce the number of plastic carrier bags being used.



- (a) (i) All plastic carrier bags are made from oil. State **one** benefit to our oil reserves if fewer plastic bags are made. [1]

.....

- (ii) Plastic bags end up in landfill sites. State **one** problem this causes. [1]

.....

- (iii) State **two** ways in which plastic bags can be a danger to animals. [2]

1.

2.

- (b) Before the introduction of charges, 350 million plastic bags were used every year. Five years later this figure had dropped to 70 million used every year.

- (i) Calculate the reduction in the number of bags used every year. [1]

reduction = million

- (ii) Calculate the percentage of bags used after five years compared with those used before charges were introduced. [2]

answer = %

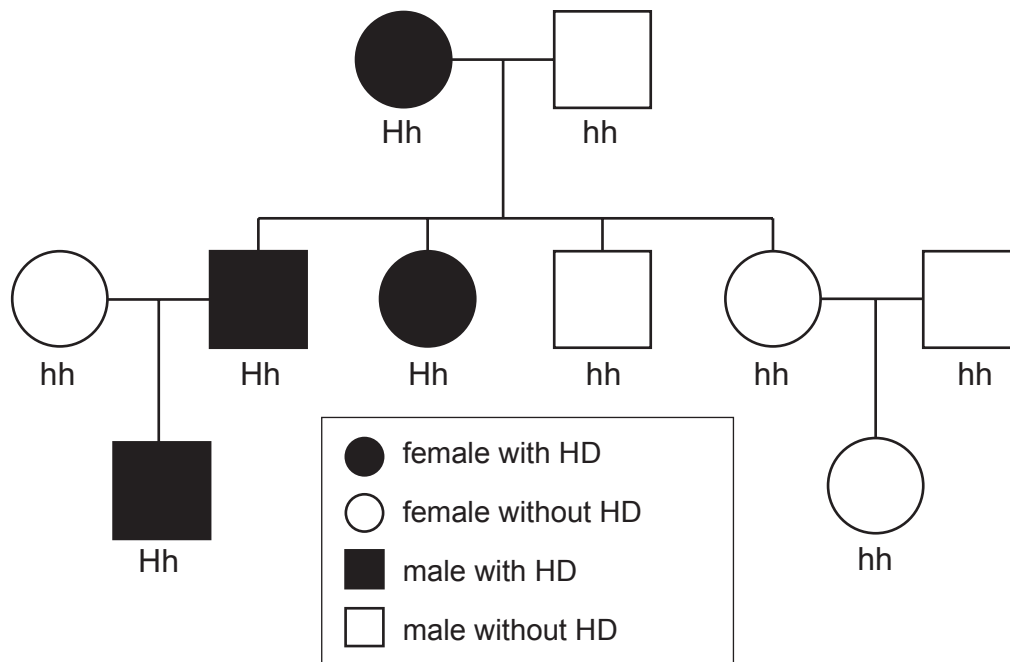
- (c) When researchers collected reused plastic bags from supermarket shoppers, they found large numbers of microorganisms in almost all carrier bags.

Underline the correct term in the brackets in the following sentences.

- (i) Microorganisms that cause disease are called (**antitoxins** / **vaccines** / **pathogens**). [1]
- (ii) The body defends itself against harmful microorganisms using (**white blood cells** / **red blood cells** / **nerve cells**). [1]

2. Huntington's disease (HD) is an inherited disease. People are born with the defective gene, but symptoms usually don't appear until middle age. Early symptoms of HD may include uncontrolled movements, clumsiness or balance problems. The disease is caused by an inherited defect in a single gene. With only one copy of the defective allele, a person can develop the disease.

The diagram shows a family tree identifying those who are unaffected (do not have HD) and those with HD.



- (a) (i) State the letter of the dominant allele. [1]
- (ii) State the genotype of a person with HD. [1]
- (b) **Complete** the Punnett squares and state the percentage chances of a child developing HD if:
- one parent does not have HD and one parent is heterozygous
 - both parents are heterozygous
- [4]

	H	h
h		
h		

% chance =

	H	h
H		
h		

% chance =

- (c) State **one** ethical problem faced by a person if they find out they have HD as a result of going for a genetic screening test. [1]

.....

.....

7

3. The diagram below shows an absorption spectrum of an object in space.



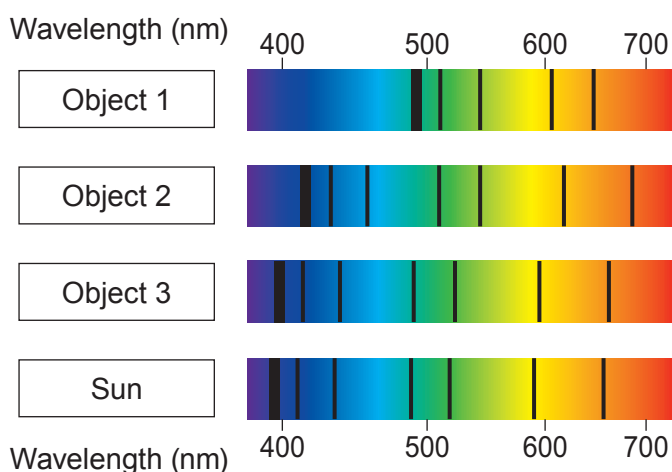
- (a) Complete the following sentences by underlining the correct word(s) in the brackets. [3]

The black lines are caused by light being (**absorbed** / **blocked** / **emitted**) by gases.

The spectrum provides information about the (**size** / **density** / **chemical makeup**) of the object.

The spectrum also provides information about how fast an object is (**expanding** / **using up its fuel** / **moving away from us**).

- (b) The diagram below shows absorption spectra from four different objects in space. The bottom spectrum is from our Sun.



Use the information in the absorption spectra above to answer the following questions. [4]

- (i) State which object is farthest away from Earth.
- (ii) State which object is moving away most slowly from Earth.
- (iii) State **one** conclusion you can make about the chemical makeup of all the objects.
.....
- (iv) State for which object the wavelength of the black lines has increased the most when compared with our Sun.
.....

4. A group of students were using dice to model radioactive decay. There were 120 dice in total.



- (a) (i) State the chance of any one of the dice landing with a 6 facing upward. [1]

.....

- (ii) Predict how many dice will land with a 6 facing up on the first throw. [1]

.....

- (iii) State what the students should do after throwing the 120 dice for the first time. [1]

.....

- (iv) Describe how they should continue the experiment. [2]

.....

.....

(b) The table shows the results.

Throw	Number of sixes	Number of dice remaining
0	0	120
1		101
2	15	86
3	14	72
4	11	61
5	11	

(i) **Complete** the table. [2]

(ii) How many dice remain after **one** half-life? [1]

(iii) Use the data in the table to estimate the half-life of the 'dice'. [1]

half-life = throws

(iv) Calculate the number of half-lives that will occur for the number of dice to fall from 120 to 15. [2]

number of half-lives =

- (c) Radioisotopes have many medical uses. One use is to sterilise blood products for transfusion.



The table below lists some radioisotopes.

Radioisotope	Half-life	Type of emissions
chromium-51	28 days	gamma
caesium-137	30 years	beta and gamma
bismuth-209	2×10^{19} years	alpha

The radioisotope used to sterilise blood products is caesium-137.

- (i) Explain why caesium-137 is a better choice than chromium-51 for this purpose. [2]

.....

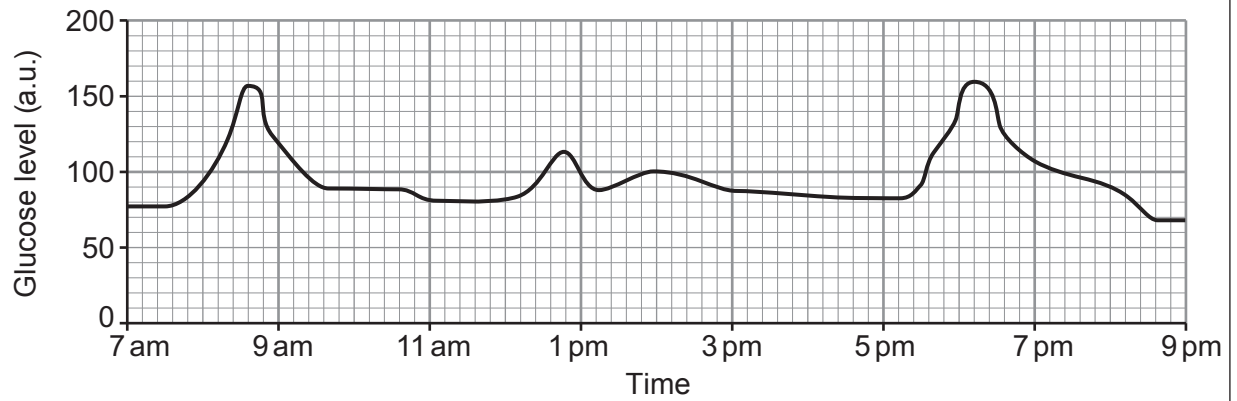
.....

- (ii) Explain why caesium-137 is a better choice than bismuth-209 for this purpose. [2]

.....

.....

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only



Use the information in the diagram and your own knowledge to explain how Nia's blood glucose is controlled. [6 QER]

- (b) Diabetics cannot control their blood glucose levels. State **two** ways in which diabetes can be treated. [2]

1.

2.

- (c) (i) State the test used to detect the presence of glucose in urine and describe how this test is carried out. [2]

test:

description:

- (ii) State what results would be observed in this test in a diabetic patient and a person not suffering from diabetes. [2]

diabetic:

non-sufferer:

Section B

*Answer **all** questions in the spaces provided.*

Use the information in the separate Resource Folder to answer the following questions.

6. (a) Use the information in **Table 1** to answer the following questions.

(i) State the value of the emissions of methane from the UK in 2005. [1]

..... MtCO₂eq

(ii) In which year did the UK emit 40 MtCO₂eq of nitrous oxide? [1]

.....

(b) Use the information in **Table 2** to answer the following questions.

(i) Calculate the increase in the concentration of methane since the pre-industrial baseline. [1]

increase =

(ii) Name the gas that has only increased because of man-made additions. [1]

.....

(c) Use the information in **Graph 3** to describe how changes in the temperature anomaly compare with changes in sun spot number and CO₂ concentrations. [2]

.....

.....

.....

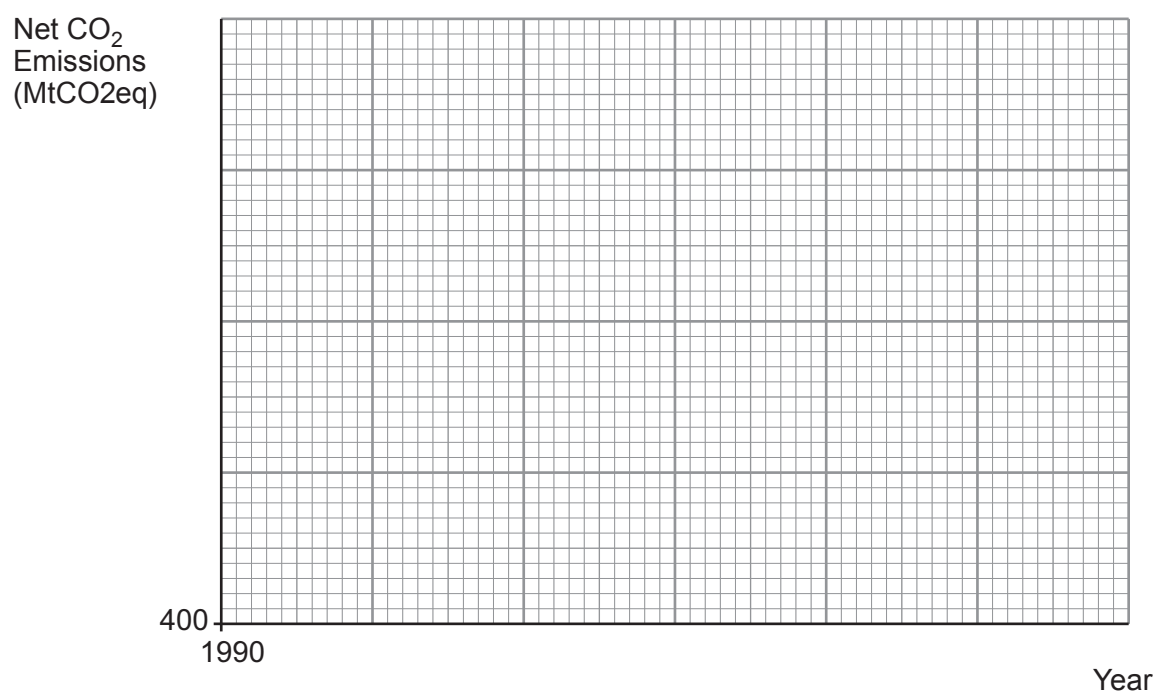
7. (a) (i) Use the information in **Table 1** to describe the trend in emissions of HFC between 1990 and 2014. [2]

.....

.....

.....

- (ii) Use the information in **Table 1** to plot a graph of net CO₂ emissions from the UK between 1990 and 2014. [4]



- (iii) Describe the trend shown in the graph. [2]

.....

.....

.....

- (iv) Describe how your graph (opposite) compares with the information in **Graph 1**. [2]

.....

.....

.....

- (v) Use your knowledge of the carbon cycle to account for the seasonal fluctuations in CO₂ concentration shown in **Graph 1**. Give reasons for your answer. [3]

.....

.....

.....

.....

- (b) Use the information in **Tables 2** and **3** and the equation:

$$\text{greenhouse contribution} = (\text{concentration in the atmosphere}) \times (\text{multiplier})$$

to complete the table below.

[3]

greenhouse contribution						
gas	multiplier	pre-industrial baseline	natural additions	man-made additions	total relative contribution	% of total contribution
carbon dioxide	1	288 000		11 880	368 400	72.37
methane	21	17 808	12 117		36 645	

- (c) Use the information in **Tables 2** and **4** to describe the effect on the data in the last column of **Table 2** when water vapour is included. [2]

.....

.....

- (d) State **one** conclusion that can be made from the information shown in **Chart 1** on page 4 of the resource folder. [1]

.....

.....

END OF PAPER

Group	
1	2
1	2
3	4
3	4
5	6
6	7
7	0

Key

Ar	Symbol	Name	Z
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Surname	Centre Number	Candidate Number
Other Names		0



GCSE – NEW

3445U20-1



APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life

FOUNDATION TIER

MONDAY, 11 JUNE 2018 – MORNING

1 hour 30 minutes

Section A	For Examiner’s use only		
	Question	Maximum Mark	Mark Awarded
	1.	10	
	2.	16	
	3.	8	
	4.	6	
	5.	10	
	Section B	6.(a)(b)	6
6.(c)(d)(e)		19	
Total		75	

3445U201
01

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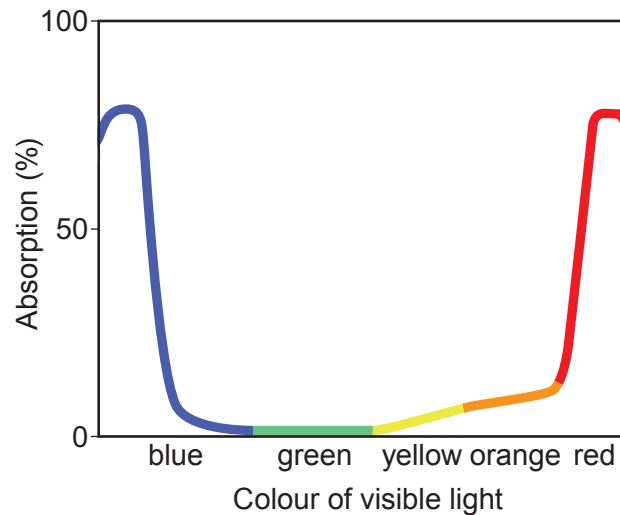
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A periodic table is printed on page 20.

Section A

Answer all questions in the spaces provided.

1. Green plants produce their own food by photosynthesis.
- (a) Green leaves capture a small percentage of solar energy. The graph below shows how the absorption of visible light by a green leaf depends on the colour of visible light.



- (i) State **one** colour of visible light that is absorbed very well.

[1]

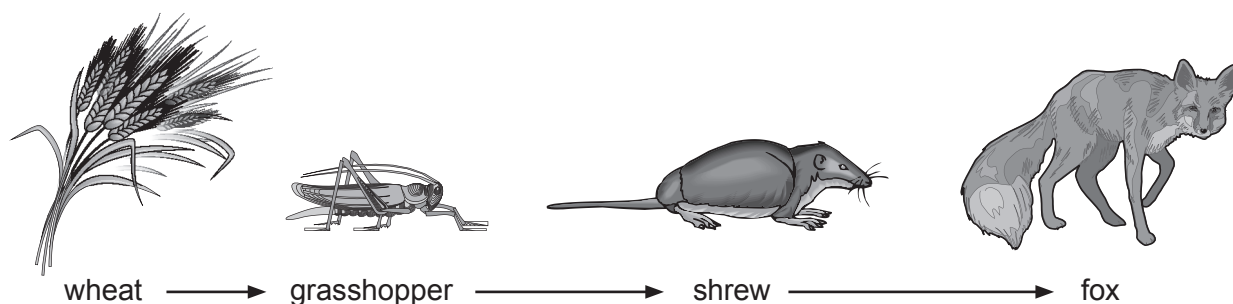
.....

- (ii) State **one** colour of visible light that is not absorbed at all.

[1]

.....

(b) The table shows what happens to the energy taken in by organisms in a food chain.



Organism	Energy per day (MJ)			
	released during respiration	used for growth	in waste products	total
wheat	10	8	6	
grasshopper	22	3	14	39
shrew	26	4		50
fox	32	4	24	60

Use the information above to answer the following questions.

(i) **Complete** the table. [2]

(ii) I. State which organism releases the greatest amount of energy during respiration. [1]

.....

II. Give **one** reason why animals respire more than plants. [1]

.....

(iii) Name the herbivore. [1]

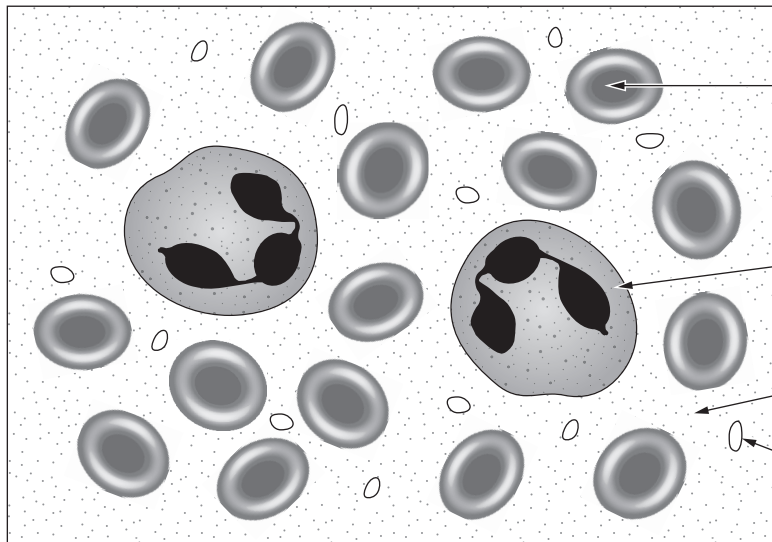
(iv) Name the prey of the shrew. [1]

(v) **Sketch** a pyramid of numbers for the food chain in the space below. [2]

2. (a) Blood consists of plasma, red blood cells, platelets and white blood cells. Each part has a different function.

(i) Label the diagram of the blood below.

[3]



A

B

C

D

(ii) Complete the table below.

[3]

Part of the blood	Function
plasma	transports carbon dioxide, soluble food and urea
red blood cells	
platelets	
white blood cells	

- (b) Gareth has blood samples taken before a cycle race. His results are shown in the table below together with normal ranges.

	Red blood cell count (million / cm ³)	White blood cell count (thousand / cm ³)	Platelets (thousand / cm ³)
<i>Normal range</i>	<i>4.4 - 5.8</i>	<i>3.9 - 10.8</i>	<i>130 - 400</i>
Gareth	2.2	6.6	105

- (i) Describe how Gareth's blood sample results compare with the normal range. [2]

.....

.....

.....

- (ii) State **one** health problem shown by Gareth's results. [1]

.....

(c) Gareth completed a training ride around a cycle track.

- (i) Gareth accelerates from rest to a velocity of 15 m/s in 2.5 seconds.
Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate his acceleration.

[2]

$$\text{Acceleration} = \dots\dots\dots \text{m/s}^2$$

- (ii) Later in the ride he travels 192 m in 12 s at a constant speed.
Use the equation:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

to calculate his speed.

[2]

$$\text{Speed} = \dots\dots\dots \text{m/s}$$

- (iii) Gareth cycles 5 laps at this constant speed. Each lap is 500 m.

I. Calculate the distance he travels.

[1]

$$\text{Distance} = \dots\dots\dots \text{m}$$

II. Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the time taken to travel this distance.

[2]

$$\text{Time} = \dots\dots\dots \text{s}$$

3. Water quality can be monitored using invertebrates as indicators.

(a) A method for an investigation to monitor water pollution using invertebrates is described below, however the steps are in the wrong order.

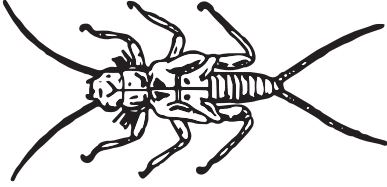
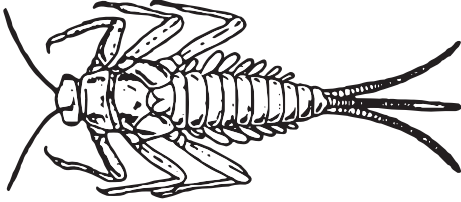
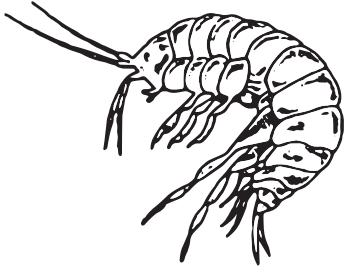
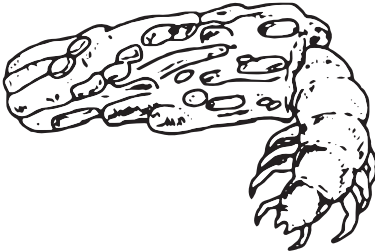
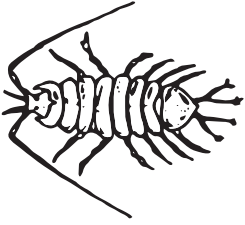
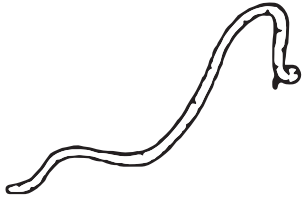
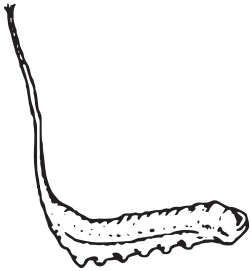
- A. Identify **two** locations to be tested e.g. two different areas in the same stream.
- B. Record the total biomass of each species of invertebrate that has been caught.
- C. Study the organisms in the tray and try to identify the invertebrates.
- D. Collect some water in a tray so it is about 2 to 3 cm deep.
- E. Collect samples of invertebrates using a net and transfer them to the tray.
- F. Pour contents gently back into the stream.

Arrange the steps in order by placing the letters **B**, **C**, **D** and **E** in the correct boxes below.

[3]

A					F
----------	-------	-------	-------	-------	----------

(b) Some invertebrate species used as indicators are shown below.

Water quality	Indicator species	
Clean water	 Stonefly nymph	 Mayfly nymph
Some pollution	 Freshwater shrimp	 Caddisfly larva
Moderate pollution	 Water louse	 Bloodworm
High pollution	 Sludgeworm	 Rat-tailed maggot
Very high pollution	No life	

(Not drawn to scale.)

The results from such an investigation are shown in the table below.

	Total biomass in sample (g)	
	Stream A	Stream B
Mayfly nymph	2	0
Caddisfly larva	55	0
Freshwater shrimp	78	2
Water louse	8	6
Bloodworm	6	22
Sludgeworm	1	110

Use the information in both tables to answer the following questions.

- (i) I. Name the **two** most common invertebrates found in stream **A**. [1]

..... and

- II. State the water quality in stream **A**. [1]

- (ii) Explain whether stream **B** has the same water quality as stream **A**. [2]

.....

- (iii) State how the results would be different for a stream with very high pollution. [1]

.....

4. Hospitals take great care to prevent patients becoming infected with bacteria such as MRSA. They often display posters such as the one shown below.

Protect patients from MRSA

The illustration is divided into three panels, each with a circular inset icon above it. In the first panel, a patient is in a hospital bed with a nurse standing by. The inset shows a pink circle with a surgical incision. In the second panel, a patient is in bed while another person lies on a gurney in the background. The inset shows a blue circle with white bacteria. In the third panel, a patient is in bed with a nurse, and a drip chamber is visible. The inset shows a circle with pills and a medicine bottle.

Surgery allows bacteria to enter the body

Bacteria can spread when infection control measures are not taken

Antibiotics save lives but poor prescribing practices put patients at risk

Use a combination of infection control measures with every patient to prevent infections

Use the information in the poster, and your own knowledge to answer the following question.

Describe effective control measures used to prevent MRSA and explain why infections such as MRSA have become difficult to treat. [6 QER]

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.....

.....

6

5. Nuclear medicine involves the use of some radioactive isotopes, that emit gamma (γ) rays, in the diagnosis and treatment of disease.

- (a) Complete the following paragraph about gamma rays using the words from the box. [4]
Each word may be used once, more than once, or not at all.

particles	low	absorb	sound	strong
light	electromagnetic	penetrate	high	

Gamma rays are frequency waves. They are used in nuclear medicine because they easily the human body. They have ionising power.

- (b) Information about some isotopes that emit gamma rays is given in the table below.

Isotope	Half-life
palladium-103	17.0 days
iodine-125	59.6 days
thallium-201	73.0 hours
cobalt-60	5.26 years

Use the information in the table to answer the following questions.

- (i) State which isotope takes the longest time to decay. [1]

.....

- (ii) State what happens to the activity of thallium-201 after 73 hours. [1]

.....

- (iii) In order to treat prostate cancer, palladium-103 pellets are placed directly into the prostate gland. They remain permanently in place.

I. Complete the table below.

[2]

Time (days)	0	17	34		68
Activity of palladium-103	1	$\frac{1}{2}$		$\frac{1}{8}$	$\frac{1}{16}$

- II. The gamma rays from the pellets are undetectable when their activity drops to $\frac{1}{32}$ of its original value. Calculate the time taken for this reduction to occur.

[2]

Time = days

10

Section B

Answer all questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

6. (a) Use the information in **Table 1** to answer the following question.

Tick (✓) the boxes next to the **three** correct statements.

[3]

Saturn orbits the Sun with a velocity of 5 km/s faster than Pluto

☐

Mercury is the hottest planet

☐

All the rocky planets have an atmosphere containing carbon dioxide

☐

Mars and the Earth have the same day length

☐

The Earth has the greatest density

☐

The Moon is the smallest planet

☐

- (b) Use the information in **Diagram 3** and your knowledge to answer the following questions.

- (i) State how the wavelength of the dark line, λ , for the very distant galaxy is different from the same line in the nearby galaxy. [1]

.....

- (ii) State how the frequency of this line for a nearby galaxy compares with the same line in a very distant galaxy. [1]

.....

- (iii) State **one** difference about the motion of the very distant galaxy compared to the nearby galaxy. [1]

.....

(c) Use the information in **Table 1** to answer the following questions.

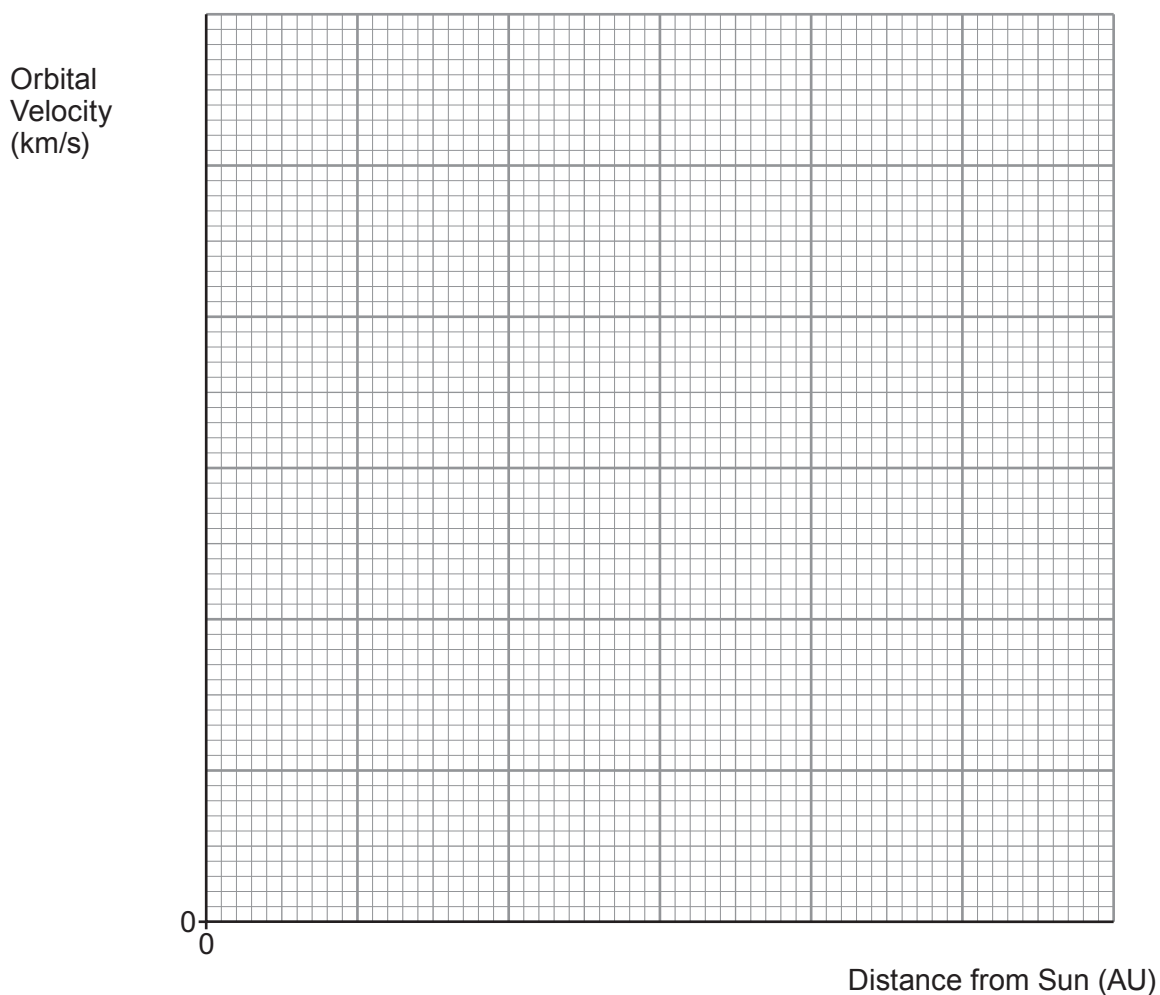
- (i) Europa is an asteroid. Estimate its temperature and orbital period. [2]

Temperature = °C

Orbital period = days

- (ii) Pluto is no longer classed as a planet. State **one** reason why. [1]

- (iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line. [4]



- (iv) Orbital velocity is not proportional to the distance from the Sun. Explain how your graph shows this to be true. [2]

(d) Use your knowledge and the information on **pages 4 and 5** to answer the following questions.

(i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

.....

.....

(ii) State **two** ways in which human activities may be affected by solar flares. [2]

1.

2.

(iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

.....

.....

.....

(e) Use the information in **Diagram 1** to describe the differences between Aristotle's model of the Solar System and the 2006 model. [4]

.....

.....

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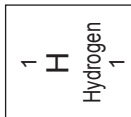
END OF PAPER

THE PERIODIC TABLE

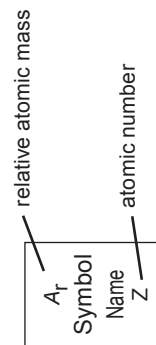
1 2 3 4 5 6 7 0

Group

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Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3445U20-1



APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life

FOUNDATION TIER

FRIDAY, 7 JUNE 2019 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
Section A	1.	6
	2.	8
	3.	6
	4.	6
	5.	7
	6.	17
Section B	7.	6
	8.	19
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Question 4 is a quality of extended response (QER) question where your writing skills will be assessed.

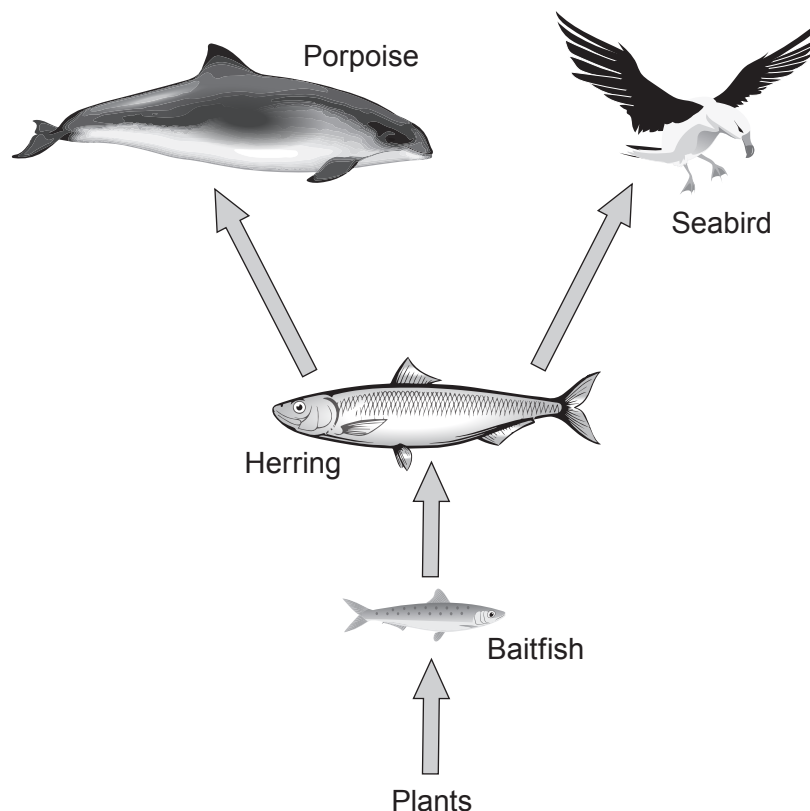
You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

Section A

Answer all questions in the spaces provided.

1. A marine food web is shown in the diagram below.



- (a) Answer the following questions about the food web.

(i) Name the producer. [1]

(ii) Name the herbivore. [1]

(iii) State **two** effects on the food web if the herring are over-fished. [2]

1.

2.

- (b) PCBs are man-made pollutants which are harmful to animals in this food web.

Explain why PCBs are more harmful to porpoises than to baitfish. [2]

.....

.....

.....

2. As the human population ages, more people depend on drug treatments.

- (a) (i) One of the stages involved in developing a new drug is computer modelling. State **three** other stages in testing a new drug. [3]

Stage 1

Stage 2

Stage 3

- (ii) State **two** ethical reasons against drug testing. [2]

1.

2.

(b) Aspirin is a widely used drug.

- (i) Aspirin acts as a painkiller. Explain **one** other use of aspirin. [2]

.....

.....

.....

- (ii) State **one** harmful side effect of taking aspirin for a long time. [1]

.....

3. The numbers of a species in a habitat can be investigated by using quadrats.



The results of an investigation into the variety of snail species found in two habitats are shown below.

Habitat X

Quadrat	Number of snails				Total number of snails
	Species A	Species B	Species C	Species D	
1	0	2	3	1	6
2	0	0	3	2	5
3	0	1	3	1	5

Habitat Y

Quadrat	Number of snails				Total number of snails
	Species A	Species B	Species C	Species D	
1	0	0	1	2	3
2	0	0	4	3	7
3	0	0	1	3	4

- (a) Use the data above to answer the following questions.

- (i) State how many species of snail are found in habitat X. [1]
- (ii) State which species are found in both habitats. [1]
- (iii) Habitat X has an area equal to 15 quadrats. Estimate the total number of snails of species C you would expect to find in habitat X. [2]

Total number =

- (iv) State **one** way the experiment could be improved so you have more confidence in your answer in (iii). [1]

.....

- (b) Snail **A** has the scientific name *Cornu aspersum*. State the advantage of using its scientific name rather than its common name. [1]

.....

6

4. State the names of the regions of the electromagnetic spectrum and describe how they can be used to take images of parts of the Universe. [6 QER]

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6

5. Humans are protected from the effects of disease by vaccination programmes and the use of antibiotics.

(a) (i) Tick (✓) **three** boxes next to the correct statements about vaccines. [3]

vaccines contain antigens

☐

vaccines contain lymphocytes

☐

vaccines stimulate the production of antibodies

☐

vaccines only protect against viruses

☐

vaccines can contain modified viruses

☐

vaccines contain antibodies

☐

- (ii) Measles is a highly infectious viral illness that is uncommon in the UK because of the effectiveness of vaccines. The measles vaccine provides life-long protection against the disease.

Flu vaccines are also available. Unlike the measles vaccine, they need to be given every year.

State **one** reason why a flu vaccine needs to be given every year. [1]

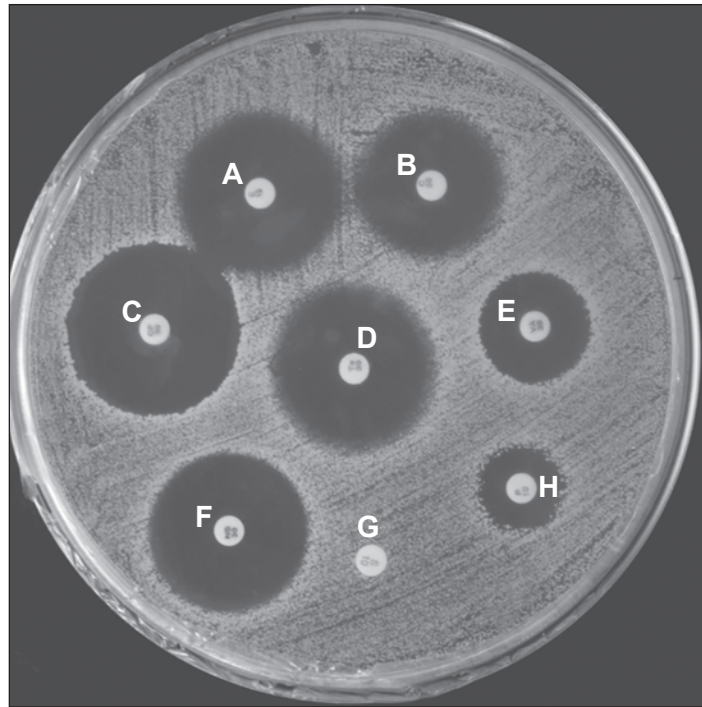
.....

.....

- (b) An antibiotic is a substance that slows down or stops bacterial growth. Not all antibiotics work against all bacteria. Therefore it is important to know which antibiotics are most effective against different bacteria.

The photograph shows the results from an experiment to determine the effectiveness of different antibiotics against one type of bacteria.

Eight different antibiotics were tested. Each disc contains one of the antibiotics (**A** to **H**).



A website states that not all antibiotics are effective against this type of bacteria. With reference to **all** antibiotics (**A-H**), explain whether you agree with the website. [3]

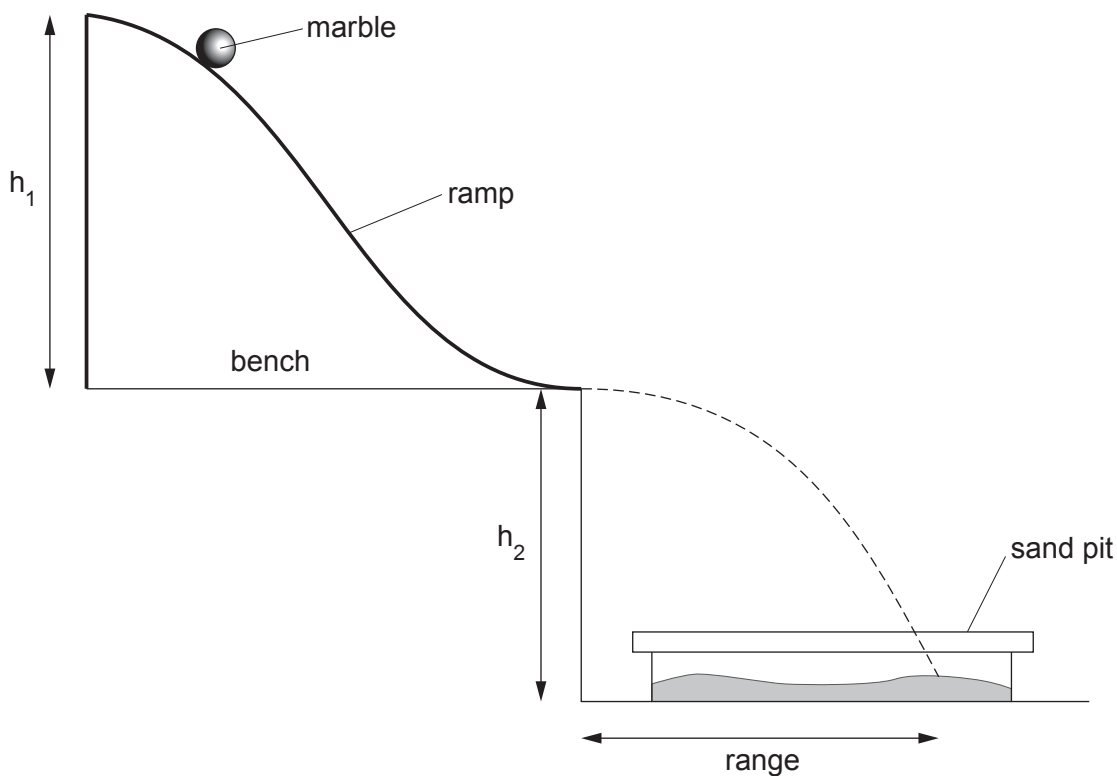
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.....

6. A model ski jump can be set up on a bench as shown in the diagram below.



The marble is released from rest at the top of the ramp. It leaves the bottom of the ramp with a certain velocity. It moves through the air and lands in the sand pit. The horizontal distance travelled in the air is called the **range**.

- (a) (i) For one height (h_1) of the ramp, the mean range was measured as 45 cm and the time the marble was in the air was 0.5 s.

Use the equation:

$$\text{velocity} = \frac{\text{mean range}}{\text{time to travel the range}}$$

to calculate the velocity of the marble as it left the ramp.

[2]

Velocity = cm/s

- (ii) The time for the marble to run down the ramp was measured 5 times. The times were 1.2 s, 1.3 s, 1.7 s, 1.2 s and 1.1 s.

- I. State which time was an anomalous result. [1]
- II. Calculate the mean time for the marble to travel down the ramp. [2]

Mean time to travel down ramp = s

- (iii) Use your answers from (i) and (ii) and the equation:

$$\text{acceleration} = \frac{\text{velocity of marble as it left the ramp}}{\text{mean time to travel down the ramp}}$$

to calculate the acceleration of the marble down the ramp. [2]

Acceleration = cm/s^2

- (b) Another experiment was carried out to find out how the acceleration of the marble depended on the height h_1 of the ramp.

- (i) The same marble was used each time. State **two** other variables that should be controlled. [2]

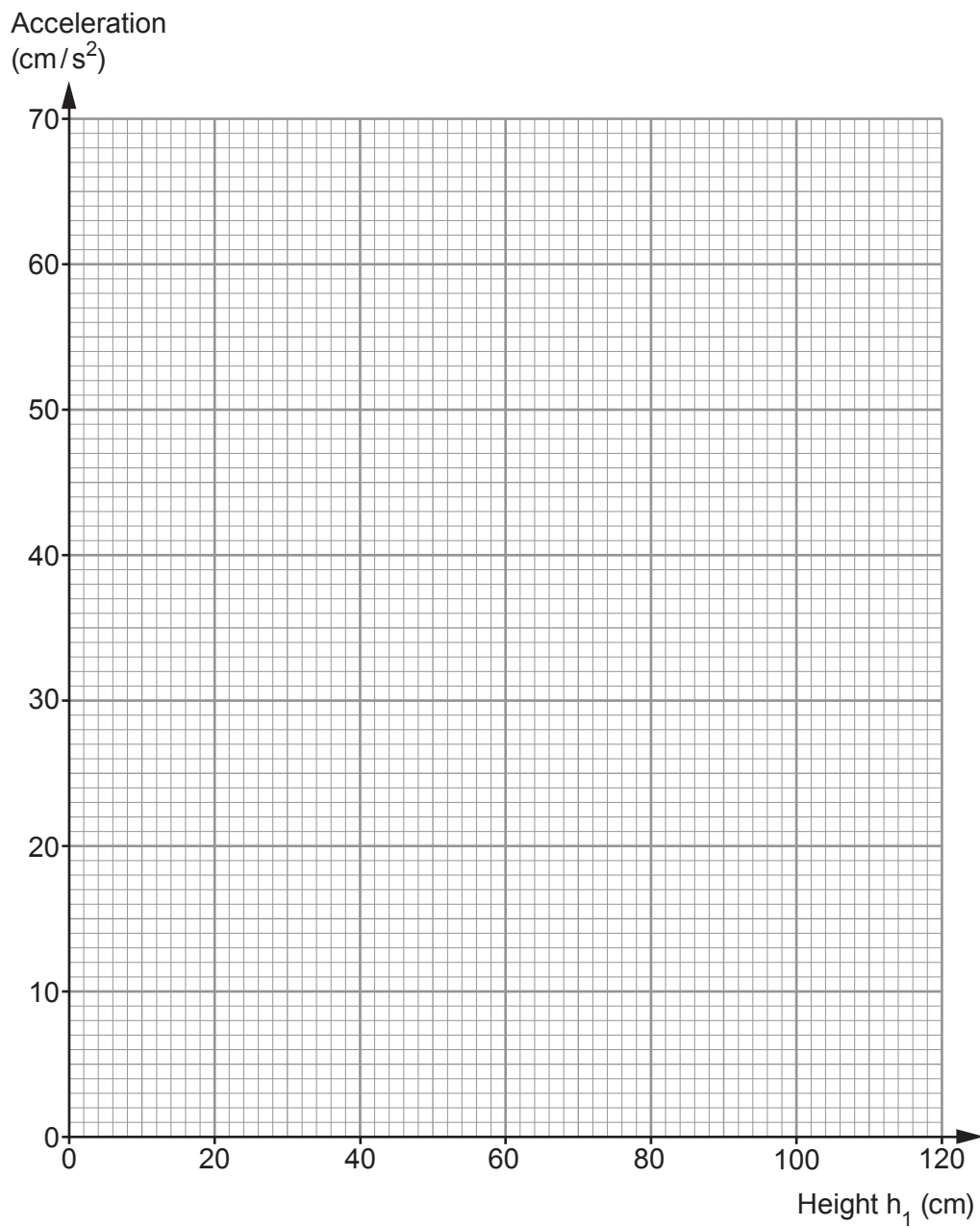
1.

2.

- (ii) The results of the experiment are shown in the table below.

Height h_1 (cm)	Acceleration (cm/s^2)
20	14
40	28
60	40
80	58
100	70

Use the data in the table to plot a graph on the grid opposite and draw a suitable line. [3]



(iii) Use the graph to find the acceleration when the height h_1 is 50 cm.

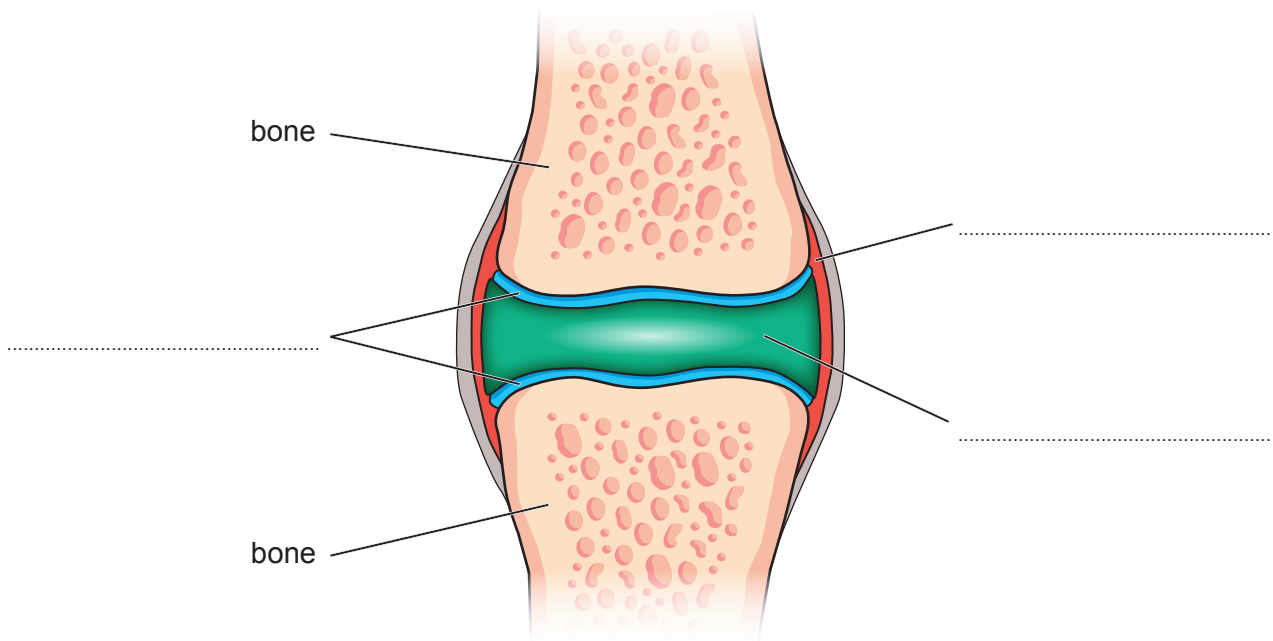
[1]

Acceleration = cm/s²

(c) Ski jumpers place their knee joints under considerable strain during their career.

- (i) A simplified diagram of a knee joint is shown below. Use words from the box to **complete** the labelling of the diagram. [3]

tendon	ligament	cartilage	synovial fluid	synovial membrane
--------	----------	-----------	----------------	-------------------



- (ii) An X-ray was taken of a ski jumper's leg following an accident.



Complete the sentence by underlining the term in the brackets.

The X-ray shows a (**simple** / **compound** / **green stick**) fracture of the leg.

Section B

Answer all questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

7. (a) Use the information in **Figure 1** to answer the following questions.

(i) State how long it takes for a person's taste to improve once they stop smoking. [1]

.....

(ii) State how long it takes for their risk of lung cancer to drop to 50% of a smoker once a person stops smoking. [1]

.....

(b) Use the information in **Figure 2** to answer the following questions.

(i) State the first year in which the percentage of smokers who quit became greater than the percentage of current smokers. [1]

.....

(ii) Give **one** reason why you cannot be certain that your answer in (b)(i) above was the actual year for this to happen. [1]

.....

.....

(iii) In 2006, a ban on smoking in the workplace and enclosed public spaces was introduced. It also became illegal to buy cigarettes under the age of 18.

It was expected that this would quickly produce a large decrease in the percentage of current smokers.

Explain whether the data supports this expectation. [2]

.....

.....

.....

8. (a) Use the information in **Figure 3** to answer the following questions.

- (i) A man drinks six 175 ml glasses of wine, three pints of lager and two 25 ml glasses of spirits. Calculate the total number of units of alcohol consumed.

[1]

Number of units =

- (ii) He stopped drinking at 11 pm. He is planning to drive at 1 pm the following day. Explain whether he can be **certain** that all units of alcohol will have been processed by this time so it will be safe for him to drive. Show your calculations.

[2]

.....

(b) Use the information in **Figures 4** and **5** to answer the following questions.

- (i) Explain why it is difficult to accurately compare the nutrition facts for the three cereals shown.

[2]

.....

- (ii) Explain which cereal you would recommend for a 7-year-old child.

[2]

.....

(c) Calculate the daily personal energy requirement (PER) for a 70 kg athlete who trains for 4 hours a day.

[3]

PER = units

(d) Use the information in **Figures 6, 7 and 8** to answer the following questions.

- (i) State the mass at which a person of height 1.7 m has a high risk of developing health problems. [1]

Mass = kg

- (ii) Explain whether you agree with the statement that 'BMI is proportional to body mass'. [2]

.....

.....

.....

- (iii) Calculate the BMI for a person of body mass 75 kg and height 1.6 m. [2]

BMI = kg/m²

- (e) (i) A woman suffers from maturity onset diabetes of the young (MODY). This is caused when one parent is heterozygous for the condition. Her partner is not diabetic. Complete the Punnett square, using **D** as the dominant allele, to show the possible genotypes of the children. [3]

- (ii) State the chance of a child being born with MODY. [1]

Chance =

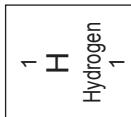
END OF PAPER

THE PERIODIC TABLE

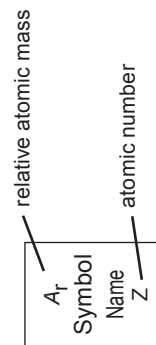
1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																		4 He Helium 2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									



Key



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445U20-1



WEDNESDAY, 15 JUNE 2022 – MORNING

APPLIED SCIENCE (Double Award)

UNIT 2: Space, Health and Life

FOUNDATION TIER

1 hour 30 minutes

Section A

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

Section B

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	4	
2.	5	
3.	7	
4.	8	
5.	10	
6.	6	
7.	10	
8.	25	
Total	75	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

You will need to refer to the separate Resource Folder to answer question **8**.

A Periodic Table is printed on page 20.



JUN223445U20101

Section A

Answer **all** questions in the spaces provided.

1. Len is responsible for monitoring the quality of water in Welsh rivers. He collects water samples from eight different sites along a river.

Diagram 1 shows the position of the collection sites.

Diagram 1

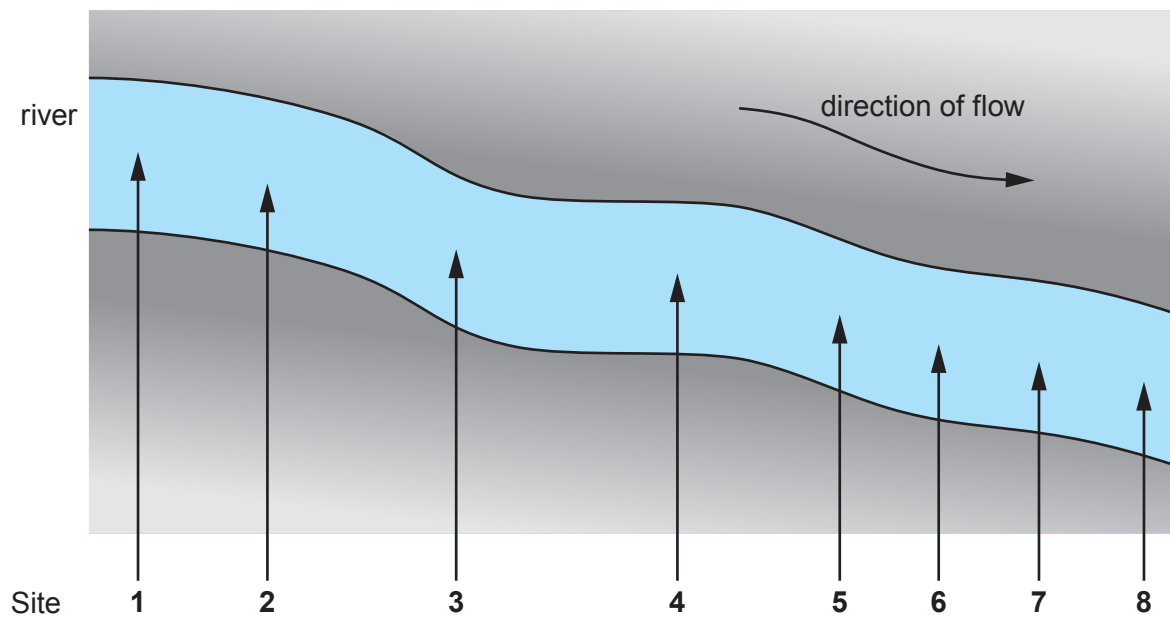
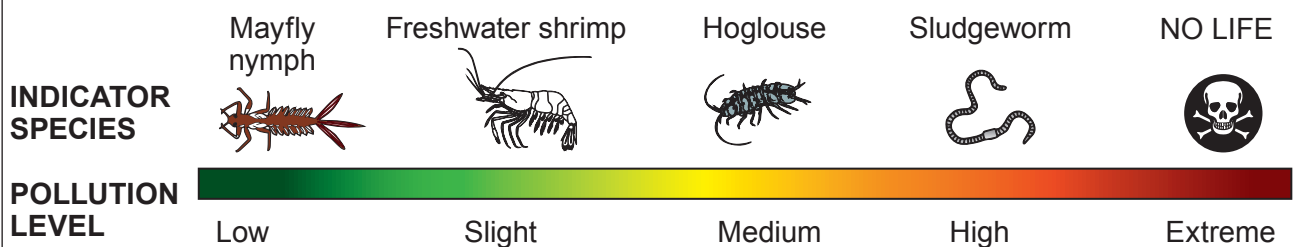


Diagram 2 shows different invertebrates and the level of water pollution in which they can live.

Diagram 2



Use the information in the diagrams to answer the following questions.

(a) Complete the table.

[3]

Site	Indicator species	Pollution level
1 & 2		low
3 & 4	No life	
5	Sludgeworm	high
6	Sludgeworm	high
7 & 8		medium

(b) Between which **two** sites did pollution enter the river?

[1]

..... and



2. Human activity is having an impact on our environment and its biodiversity.

(a) State why using more and more land for house building is reducing biodiversity. [1]

.....

(b) Protected areas such as nature reserves are an attempt to maintain biodiversity.

Tick (✓) two other methods being used to maintain biodiversity. [2]

seed banks

☐

captive breeding

☐

fishing

☐

photography

☐

(c) Explain why there need to be wildlife corridors between nature reserves. [2]

.....

.....

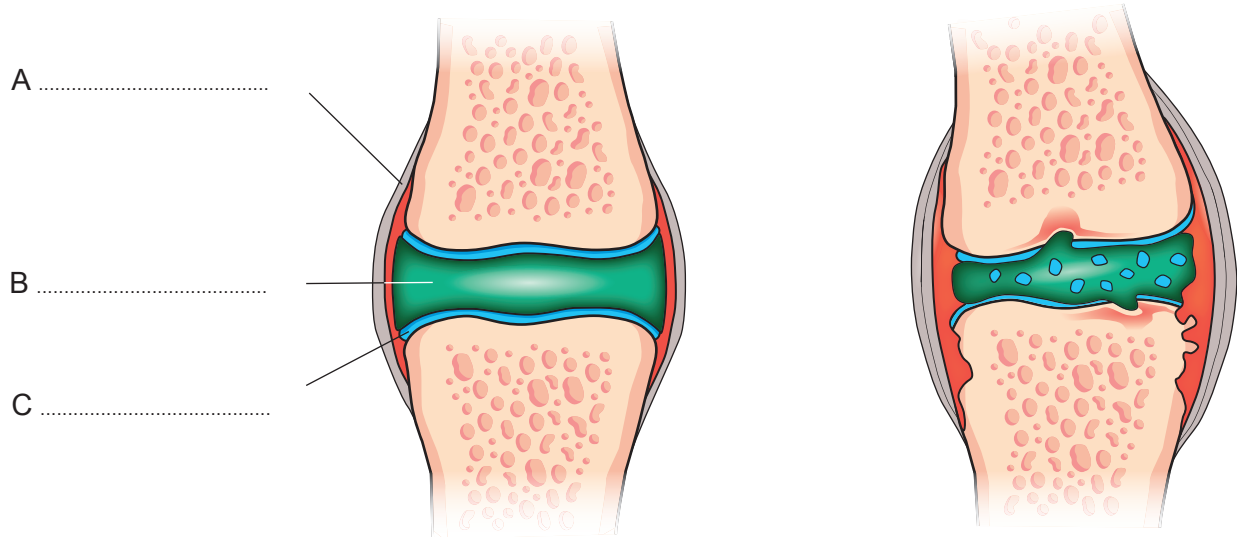
.....



3. The upper and lower leg are connected by the knee joint.

The lower leg is connected to the foot by the ankle joint.

- (a) The knee is a synovial hinge joint.
The diagram shows a healthy knee joint and one damaged by osteoarthritis.



- (i) Label the diagram using only the words from the box.

[3]

synovial fluid

ligament

muscle

cartilage

- (ii) I. State **one** cause of osteoarthritis.

[1]

.....

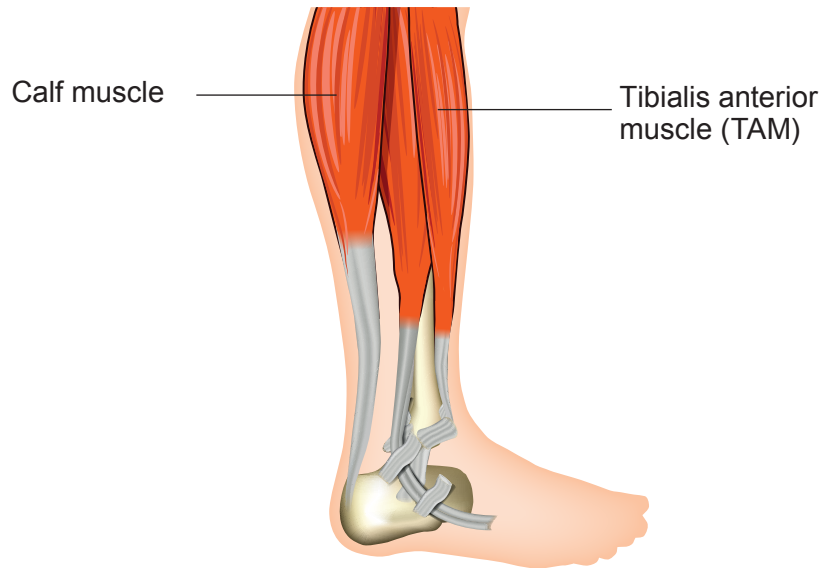
- II. State why a person suffering from osteoarthritis has knee pain.

[1]

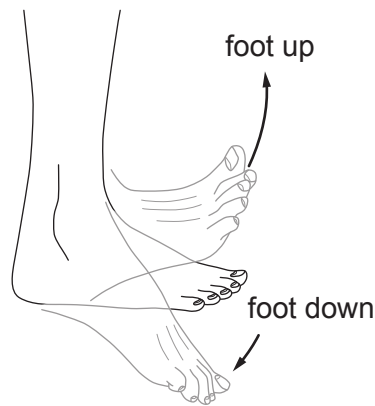
.....



- (b) The ankle joint is controlled by a pair of antagonistic muscles. These are the calf muscle and the tibialis anterior muscle (TAM).



Explain how these muscles work together to move the foot up and down as shown in the diagram. [2]



.....

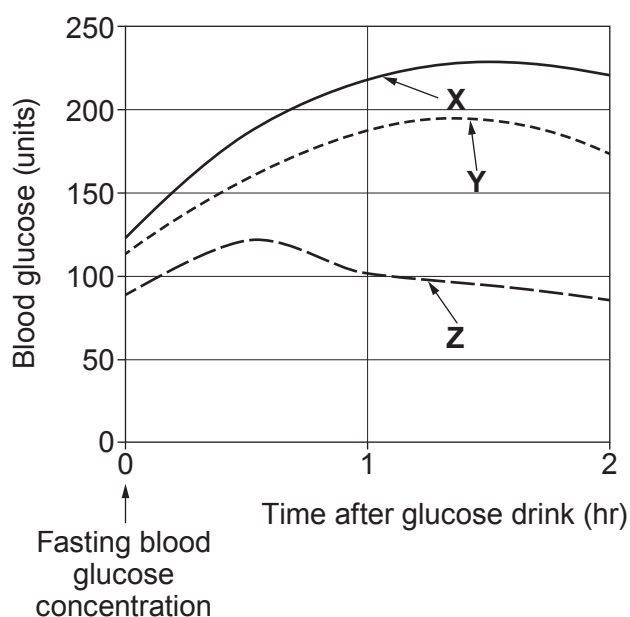
.....

.....



4. Medical scientists have developed methods to diagnose, monitor and treat diseases.

- (a) For one type of diabetes test, you fast (don't eat) overnight, and the fasting blood glucose level is measured. Then you drink a glucose drink, and blood glucose levels are tested for the next two hours. A blood glucose level less than 140 units is normal. A reading of more than 200 units after two hours indicates diabetes. A reading between 140 and 199 units indicates prediabetes.



(i) State which line, **X**, **Y** or **Z** would show a person is diabetic. [1]

(ii) State **one** method used to control diabetes. [1]

.....



- (b) Ionising radiation and ultrasound are used by medical scientists. Examples of imaging techniques are shown below. Complete the table. [4]

	Imaging technique	Example of its medical use
A		X-ray detects broken bones
B		
C		

- (c) Damage to DNA can lead to cancer.

- (i) Complete the sentence below to show the base pairs found in DNA. [1]

T pairs with and G pairs with

- (ii) Give **one** reason why imaging technique **A** could cause DNA damage. [1]

.....



5. Three images have been combined below to show colliding galaxy clusters 5 billion light years from Earth.

- The background is a Hubble Space Telescope image taken using visible light.
- The blue area is an X-ray image from the Chandra Space Observatory.
- The red area is an image from a radio telescope.



- (a) (i) State **one** advantage of using space-based telescopes to produce images of objects in space. [1]

.....

.....

- (ii) State **one** disadvantage of using space-based telescopes to produce images of objects in space. [1]

.....

.....

- (b) Complete the following sentence by underlining the correct word in the brackets.

It takes radio waves 5 billion (**seconds / hours / days / years**) to travel from the colliding galaxy clusters to reach the Earth. [1]



(c) The image is taken using X-rays, radio waves and visible light.

(i) Which of these waves has the highest frequency? [1]

(ii) Which of these waves has the longest wavelength? [1]

(iii) Name **two** other types of electromagnetic waves that could be used to take images of galaxies. [2]

..... and

(d) The speed of electromagnetic waves through space is 300 000 000 m/s. Some radio telescopes on Earth detect radio waves which have wavelengths between 2 and 10 m.

Use the equation

$$\text{frequency} = \frac{\text{wave speed}}{\text{wavelength}}$$

to calculate the **maximum** frequency of radio waves detected. [2]

frequency = Hz

(e) Images of the Universe can be used to support the current theory of the origin of the Universe.

State the name of this theory. [1]

.....



- | Nutrient | Quantity per day | |
|---------------|------------------|-------------|
| | Balanced diet | Actual diet |
| Energy | 2000 kcal | 2500 kcal |
| Protein | 50 g | 50 g |
| Fat | 70 g | 120 g |
| Carbohydrates | 310 g | 400 g |
| Sugar | 90 g | 180 g |
| Sodium (salt) | 2.3 g | 5.6 g |
| Fibre | 30 g | 9 g |

7. Tom competes in a 400 m training race. His split times to travel each 100 m are measured. His pulse rate is also measured in beats per minute (bpm) before the race starts and at every 100 m interval. The measurements are shown in **Table 1** below.

Table 1

Distance (m)	0	100	200	300	400
Time (s)	0	13	27	43	58
Pulse rate (bpm)	78	133	132	134	133

Use the data in the table and your knowledge to answer the following questions.

- (a) (i) State Tom's resting pulse rate. bpm [1]
- (ii) One minute after the race ended, Tom's pulse rate had dropped to 122.

Calculate **how much longer** it will be for his pulse to return to his resting rate, assuming it continues to drop at the same rate. [3]

time = minutes

- (iii) Use the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

and the information in **Table 1** to calculate Tom's mean speed over the race. [2]

mean speed = m/s



- (iv) Use the information in **Table 1** to complete **Table 2** to show the order of running each 100 m section from fastest to slowest. The fastest 100 m section has already been filled in. [2]

Table 2

Fastest section



Slowest section

0 to 100 m
..... to m
..... to m
..... to m

- (b) Tom then trains for 6 months. **Complete Table 3** below by estimating Tom's pulse rate every 100 m in a 400 m race after training. [2]

Table 3

Distance (m)	0	100	200	300	400
Pulse rate (bpm)					



Section B

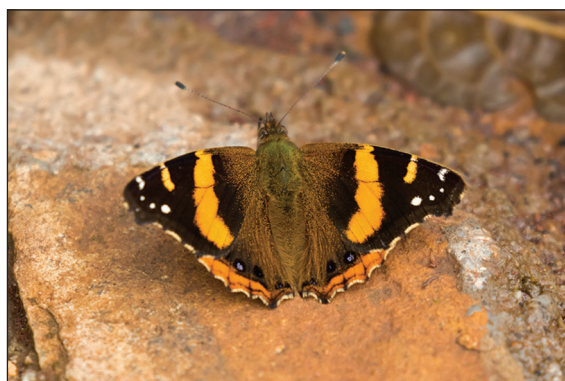
Refer to the separate Resource Folder to answer question 8.

8. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrtosiphon pisum*.

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	
Phylum:	
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	
Species:	
Scientific name:	<i>Acyrtosiphon pisum</i>

- (ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica*.



- I. State **one** advantage of using scientific names. [1]

.....

.....

- II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

.....

.....



(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

- (i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

.....

II. State **one** reason why this seasonal movement occurs. [1]

.....

.....

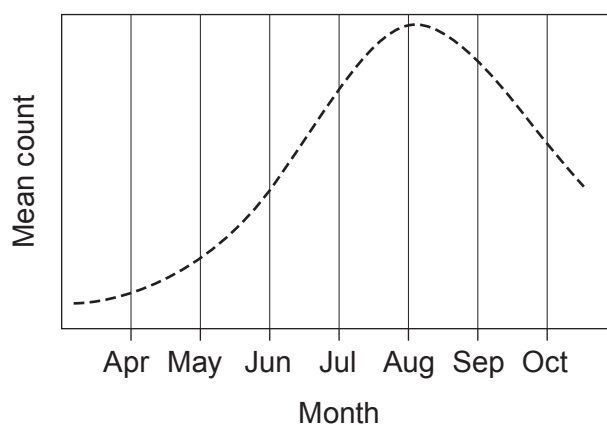
- (ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

.....

.....

.....

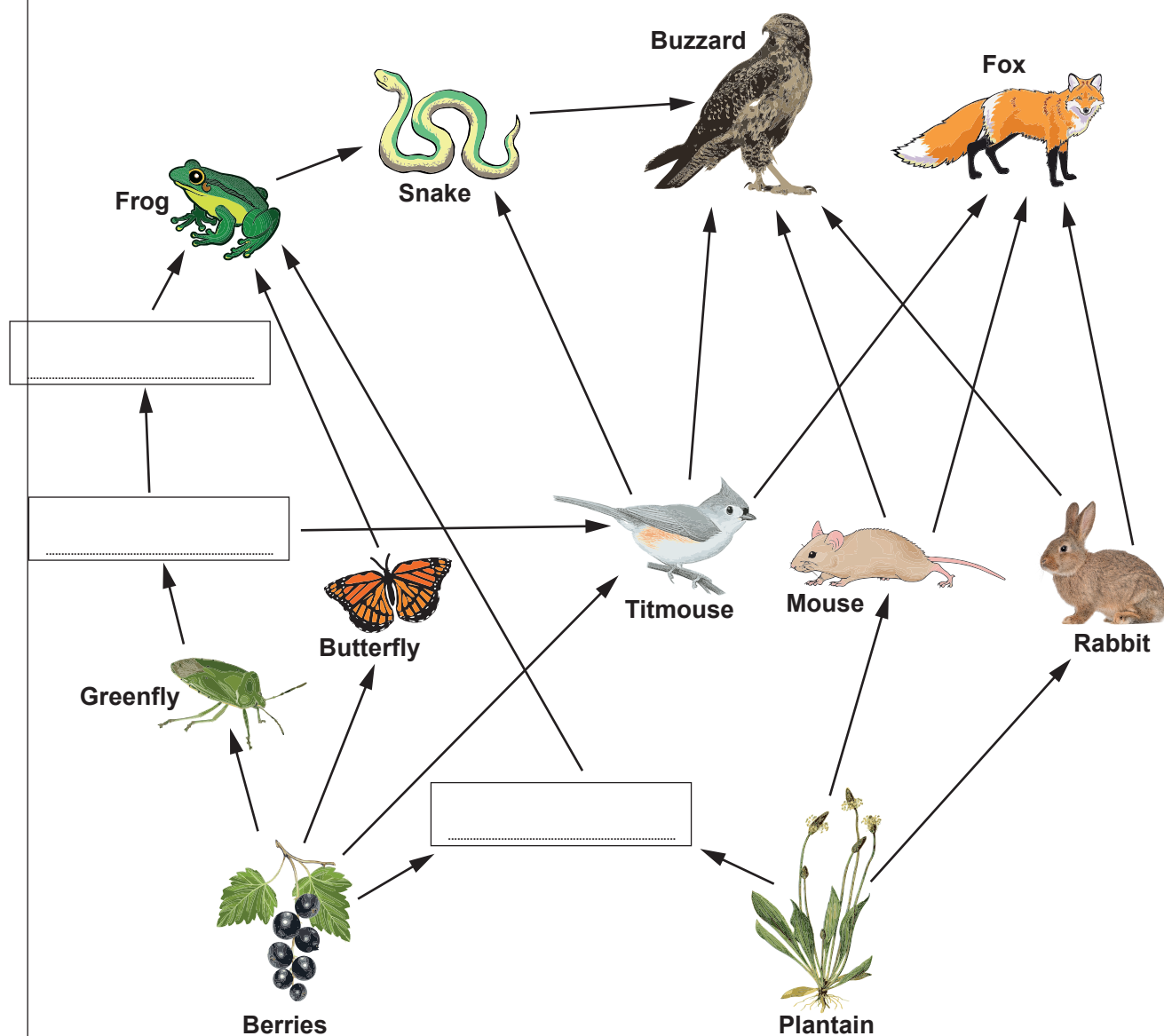
- (iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



(c) **Table 2** gives information about the feeding relationships in one woodland habitat.

(i) Use the information in **Table 2** to **complete** the food web below.

[3]



(ii) In this habitat, the fields containing plantain are cleared to make way for housing. Explain how this affects the foxes.

[3]

.....

.....

.....

.....



(iii) Explain why the frog can be found at more than one trophic level. [2]

.....

.....

.....

(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency =

END OF PAPER

8





THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0

1 H Hydrogen 1 11 B Boron 5 12 C Carbon 6 14 N Nitrogen 7 16 O Oxygen 8 19 F Fluorine 9 20 Ne Neon 10 4 He Helium 2																	
7 Li Lithium 3	9 Be Beryllium 4																
23 Na Sodium 11	24 Mg Magnesium 12																
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

relative atomic mass

Ar

Symbol

Name

Z

atomic number



3445U20-1

TUESDAY, 13 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)

**UNIT 2: Space, Health and Life
FOUNDATION TIER**

**1 hour 30 minutes plus your additional
time allowance**

Surname _____

First name(s) _____

Centre Number _____

Candidate Number 0 _____

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pen or pencil and a ruler.

ITEMS INCLUDED WITH QUESTION PAPER

A separate Data Booklet.

A separate Diagram Booklet.

The Diagram Booklet MUST be handed in to the invigilators and sent for marking.

INSTRUCTIONS TO CANDIDATES

Use black ink, black ball-point pen or your usual method.

Write your name, centre number and candidate number in the spaces provided on the front cover.

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(a) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

You will need to refer to the separate resource folder to answer questions 7, 8 AND 9.

The Periodic Table is printed in the separate Data Booklet.

(Turn over)

Section A

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	11	
3.	4	
4.	10	
5.	12	
6.	7	
7.	6	
8.	11	
9.	8	
Total	75	

Section B

SECTION A

Answer ALL questions.

- 1 Look at DIAGRAM 1.1 in the seperate diagram booklet. Our solar system contains several planets and other objects that orbit the Sun.**

Answer the following questions using your knowledge, DIAGRAM 1.1 and the words in the box below.

Each word can be used once, more than once, or not at all.

Comet

Jupiter

Mars

Oort

Venus

Pluto

(Turn over)

1 (a) Name the second rocky planet from the Sun. [1 mark]

(b) Name the gas planet closest to the Sun. [1 mark]

(c) Name the type of object that orbits the Sun in a highly elliptical orbit. [1 mark]

(Turn over)

(d) Between which TWO planets is the asteroid belt found? [2 marks]

and

1 (e) Name the cloud. [1 mark]

6

(Turn over)

2 **DIAGRAM 2.1** in the separate diagram booklet shows the carbon cycle.

(a) Use the diagram and your knowledge to answer the following questions.

(i) **CIRCLE** the name of the process shown by label **A**. [1 mark]

feeding

photosynthesis

respiration

(Turn over)

2 (a)(ii)

TICK (✓) the boxes next to the THREE correct statements. [3 marks]

- ☐ **Photosynthesis uses carbon dioxide from the atmosphere.**
- ☐ **Decomposing dead plants adds carbon dioxide to the atmosphere.**
- ☐ **Fossilising dead plants adds carbon dioxide to the atmosphere.**
- ☐ **Respiration adds carbon dioxide to the atmosphere.**
- ☐ **Burning dead plants removes carbon dioxide from the atmosphere.**
- ☐ **When green plants feed they add carbon dioxide to the atmosphere.**

(Turn over)

2 (a)(iii)

State TWO ways that human activity has caused carbon dioxide levels to increase. [2 marks]

1. _____

2. _____

(Turn over)

2 (b) TABLE 2.2 in the separate diagram booklet is taken from a climate change website.

It gives the mass of carbon dioxide produced by a person travelling to work either by car or bus.

FOUR people each travel a total of 20 km per day.

Use the information to answer the questions on the following page.

(Turn over)

$$\text{total mass of carbon dioxide produced (kg)} = \frac{\text{number of people}}{\text{mass of carbon dioxide produced per person}}$$

2 (b)(i)

Use the equation on the opposite page to answer the following questions.

- I. Calculate the total mass of carbon dioxide produced when four people drive 20 km by car. [2 marks]**

mass of carbon dioxide produced = _____ kg

(Turn over)

2 (b)(i) continued

II. Calculate the mass of carbon dioxide produced if four people travelled 20 km by bus. [1 mark]

mass of carbon dioxide produced = _____ kg

(Turn over)

2 (b)(ii)

Calculate the savings in carbon dioxide production per week if they travel by bus on the same 20 km journey from Monday to Friday instead of using their cars. [2 marks]

savings in carbon dioxide production = _____ kg

- 3 Lichen is an indicator species for air pollution.**

Students measured the abundance of lichen at various distances from the city centre.

Their results are shown in GRAPH 3.1 in the separate diagram booklet.

- (a) Use GRAPH 3.1 to answer the following questions.**
- (i) Estimate the abundance of lichen 4.0 km from the city centre. [1 mark]**

abundance = _____ units

(Turn over)

3 (a)(ii)

Describe the trend shown by the graph. [1 mark]

(b) Explain why the abundance of lichen is not the same at all the distances shown. [2 marks]

continue answer on the next page

(Turn over)

4

- 4** **DIAGRAM 4.1** in the separate diagram booklet shows the apparatus used to find the acceleration of a squash ball as it rolls down a ramp.

A stopwatch is also required.

- (a)** **Describe the method used in this investigation. [6 marks QER]**

Include in your answer:

- **How you would carry out the experiment**
 - **How to reduce the uncertainty in your results.**
-
-

continue answer on the next page

(Turn over)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(Turn over)

4 (b) The following readings were taken during the investigation.

**Time to travel 50 cm along the bench
= 2.0 s**

**Time to reach the bottom of the ramp
= 2.5 s**

(i) Use the equation:

$$\text{velocity} = \frac{0.5}{\text{time to travel 50 cm along the bench}}$$

to calculate the velocity of the ball travelling along the bench. [2 marks]

continue answer on the next page

(Turn over)

velocity = _____ m/s

4 (b)(ii) Use the equation:

$$\text{acceleration} = \frac{\text{velocity along the bench}}{\text{time to reach the bottom of the ramp}}$$

to calculate the acceleration of the ball down the ramp. [2 marks]

continue answer on the next page

(Turn over)

acceleration = _____ m/s²

10

(Turn over)

- 5 Cystic fibrosis is an example of an inherited disease. A carrier of cystic fibrosis has one unaffected allele (F) and one affected allele (f).**
- (a) Two parents are planning to have a child. The father is a carrier of cystic fibrosis (Ff), the mother is unaffected.**
- (i) CIRCLE the genotype of the unaffected mother. [1 mark]**

FF**ff****Ff**

- (ii) Complete PUNNETT SQUARE 5.1 in the separate diagram booklet to show the possible genotypes of the children. [2 marks]**

(Turn over)

5 (a)(iii)

State the chance of a child from these parents being born with cystic fibrosis. [1 mark]

(iv) The possible genotypes of children from two carrier parents are shown in

PUNNETT SQUARE 5.2 in the separate diagram booklet.

These carrier parents thought they would have the same chance of having a child with cystic fibrosis as the parents in (a)(ii). Explain whether you agree. [1 mark]

continue answer on the next page

(Turn over)

5 (b) Cystic fibrosis causes a build-up of thick mucus in the lungs. Mucus in the lungs helps bacteria to grow.

UNDERLINE TWO statements below which show ways white blood cells help to defend the body against bacterial infection. [2 marks]

Carry oxygen to the bacteria

Ingest bacteria

Produce antibodies

Make glucose

(Turn over)

5 (c)(i)

Complete TABLE 5.3 in the separate diagram booklet to show how the number of bacteria doubles every 15 minutes. [2 marks]

(c)(ii)

Antibiotic resistance is a concern. The list below describes the stages involved in the development of bacteria with antibiotic resistance. It is not in the correct order.

- 1. Many more bacteria survive antibiotic treatment.**
- 2. There was a mutation to a gene in a few bacteria.**
- 3. The survivors reproduce, passing on the mutated gene.**
- 4. Bacteria with the mutation survive antibiotics.**

(Turn over)

5 (c)(ii) continued

Arrange the stages in the correct order starting with stage 2. [2 marks]

2 **→** _____ **→** _____ **→** _____

(iii) Complete the following sentence by UNDERLINING the correct word in brackets. [1 mark]

The development of antibiotic resistant bacteria is an example of (evolution / vaccination / digestion).

- 6** Some students investigated the number of dandelion plants on a lawn. **DIAGRAM 6.1** in the separate diagram booklet shows the lawn and the location of 5 quadrats (**A to E**) which the students had placed at random.

The students counted the number of dandelions in each quadrat and recorded their results in **TABLE 6.2** in the separate diagram booklet.

- (a)** Calculate the mean number of dandelions per quadrat. [1 mark]

mean number
of dandelions = _____

(Turn over)

**6 (b) Calculate the area of the lawn.
[1 mark]**

area of lawn = _____ m²

**(c) Each quadrat has an area of 1 m².
Calculate how many quadrats will fit
in the area of the lawn. [1 mark]**

number of quadrats = _____

**(d) Use your answers to (a) and (c)
to calculate the total number of
dandelions on the lawn. [1 mark]**

number of dandelions = _____

(Turn over)

6 (e) The actual number of dandelions on the lawn is 600.

Use the formula below, and your answer to (d), to calculate the percentage error of your answer to (d). [2 marks]

$$\text{percentage error} = \frac{\text{number of dandelions} - 600}{600} \times 100$$

percentage error = _____ %

(Turn over)

- 6 (f) State ONE improvement to the method used by the students which would give more confidence in their answer for the number of dandelions. [1 mark]**

7

SECTION B

Refer to the separate Resource Folder to answer questions 7, 8 and 9.

7 X-rays are commonly used to diagnose illness.

(a) Refer to FIGURE 2 to answer the following questions.

(i) State the range of wavelength for X-rays. [1 mark]

_____ **m**

(ii) Estimate the highest frequency of X-rays. [1 mark]

_____ **Hz**

(Turn over)

7 (b) Refer to FIGURE 3 to answer the following questions.

- (i) A patient has received a total radiation dose of 70 units from X-rays during the year.**

Calculate how many chest X-rays this dose is equivalent to. [2 marks]

number of chest X-rays = _____

(Turn over)

b (ii) A patient has an X-ray taken of their spine. Calculate the fraction of the yearly background radiation this dose is equivalent to. [2 marks]

(1 year = 365 days)

fraction = _____

6

(Turn over)

8 (a) Use the information under FIGURE 1 to answer the following question.

The wave speed of ultrasound in body tissue is 1540 m/s.

Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

**to calculate the wavelength of the ultrasound used in a 2 MHz scan.
[3 marks]**

wavelength = _____ m

(Turn over)

8 (b) Use the information in FIGURE 4 to answer the following questions.

- (i) Select the most suitable radioisotope of iodine from the table in FIGURE 4 that would be used as a tracer. Explain your choice. [2 marks]**

(Turn over)

8 (b)(ii)

DIAGRAM 8.1 in the separate diagram booklet shows a patient undergoing external beam radiation therapy for a brain tumour.

Select the most suitable radioisotope from the table in FIGURE 4 that would be used for this treatment. Explain your choice. [3 marks]

(Turn over)

8 (c) Refer to FIGURES 4 AND 6 in the Resource Folder to answer the following question.

State which of the radioisotopes R1, R2, R3, R4, R5 or R6, is lutetium-177.

Explain your reasoning. [3 marks]

Radioisotope _____ is lutetium-177.

continue answer on the next page

(Turn over)

11

- 9 The following questions are based on the information on page 5 of the Resource Folder about using dice to model half-life.**
- (a) Each group's results were added together to give class results. Give ONE reason why this makes the data more repeatable. [1 mark]**

(Turn over)

- (b) The results in FIGURES 7 AND 8 are used to plot graphs on GRAPH 9.1 in the separate diagram booklet.**
- (i) Add lines to GRAPH 9.1 to determine the half-life represented by line 3. [2 marks]**

half-life = _____ throws

(Turn over)

9 (b)(ii)

Use the information in GRAPH 9.1 to tick (✓) the boxes next to the THREE correct statements. [3 marks]

☐

Line 1 represents the data from Model 2.

☐

Line 2 shows decay of the quickest rate.

☐

Line 2 shows the equivalent of three half-lives after 5 throws.

☐

Line 1 represents the longest half-life.

☐

Line 3 would have been produced if 600 coins had been used instead of dice and the number of 'heads' counted after each throw.

☐

Line 3 has an activity that is $\frac{1}{8}$ of the original after 8 throws.

(Turn over)

9 (c) The teacher suggests to the class that they add 5 red wooden cubes to the 50 dice they start with. The red cubes would not be removed after each throw but would be counted every time to improve the modelling of radioactive decay.

**(i) State what the red cubes represent.
[1 mark]**

(Turn over)

9 (c) (ii)

State how the presence of the red cubes would affect line 1 on GRAPH 9.1. [1 mark]

8

END OF PAPER



GCSE

3445U20-1

TUESDAY, 13 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)

**UNIT 2: Space, Health and Life
FOUNDATION TIER**

**1 hour 30 minutes plus your additional
time allowance**

DIAGRAM BOOKLET

**This Diagram Booklet MUST be handed in
to the invigilators and sent for marking.**

Surname _____

First name(s) _____

Centre Number _____

Candidate Number 0 _____

DIAGRAM 1.1

Not drawn to scale

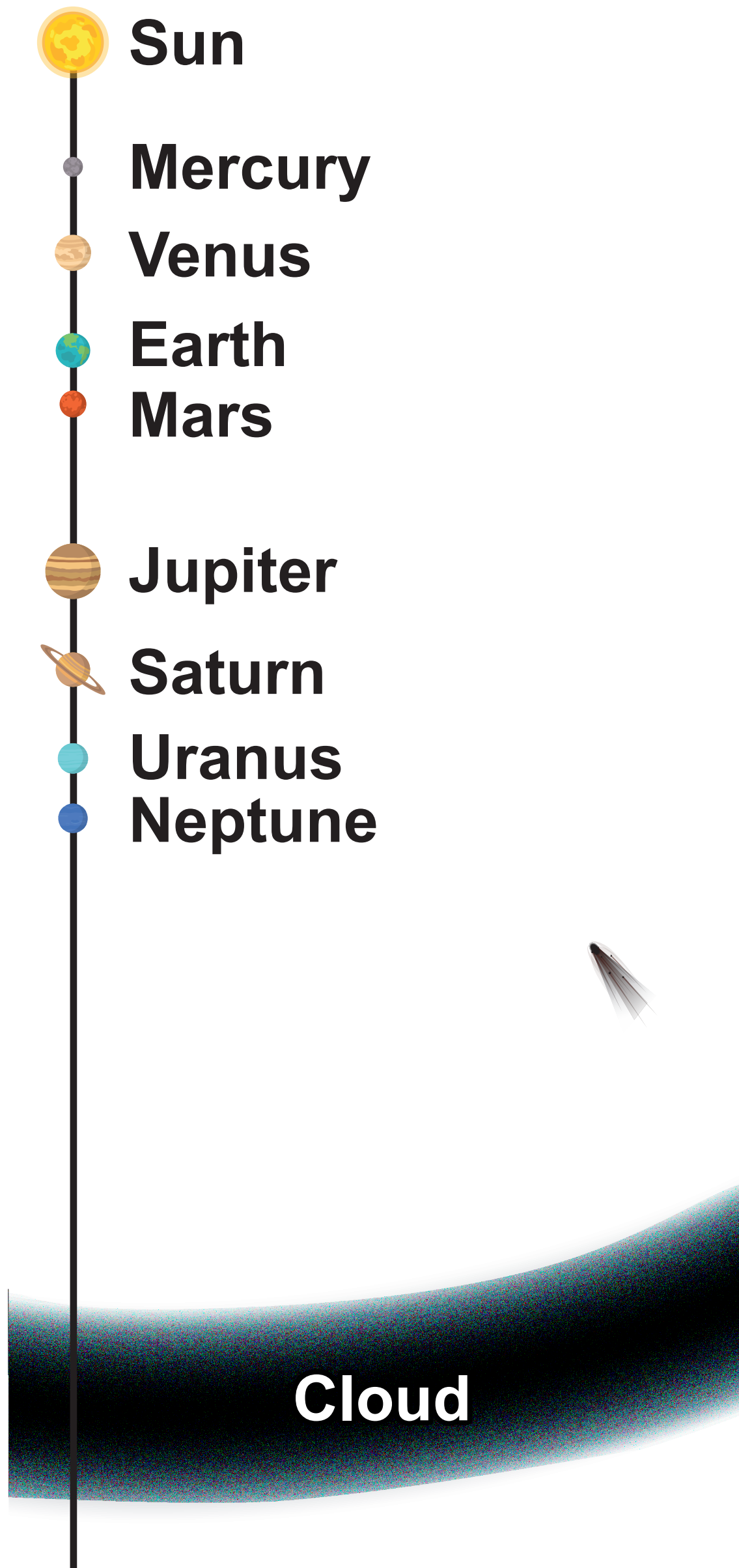


DIAGRAM 2.1

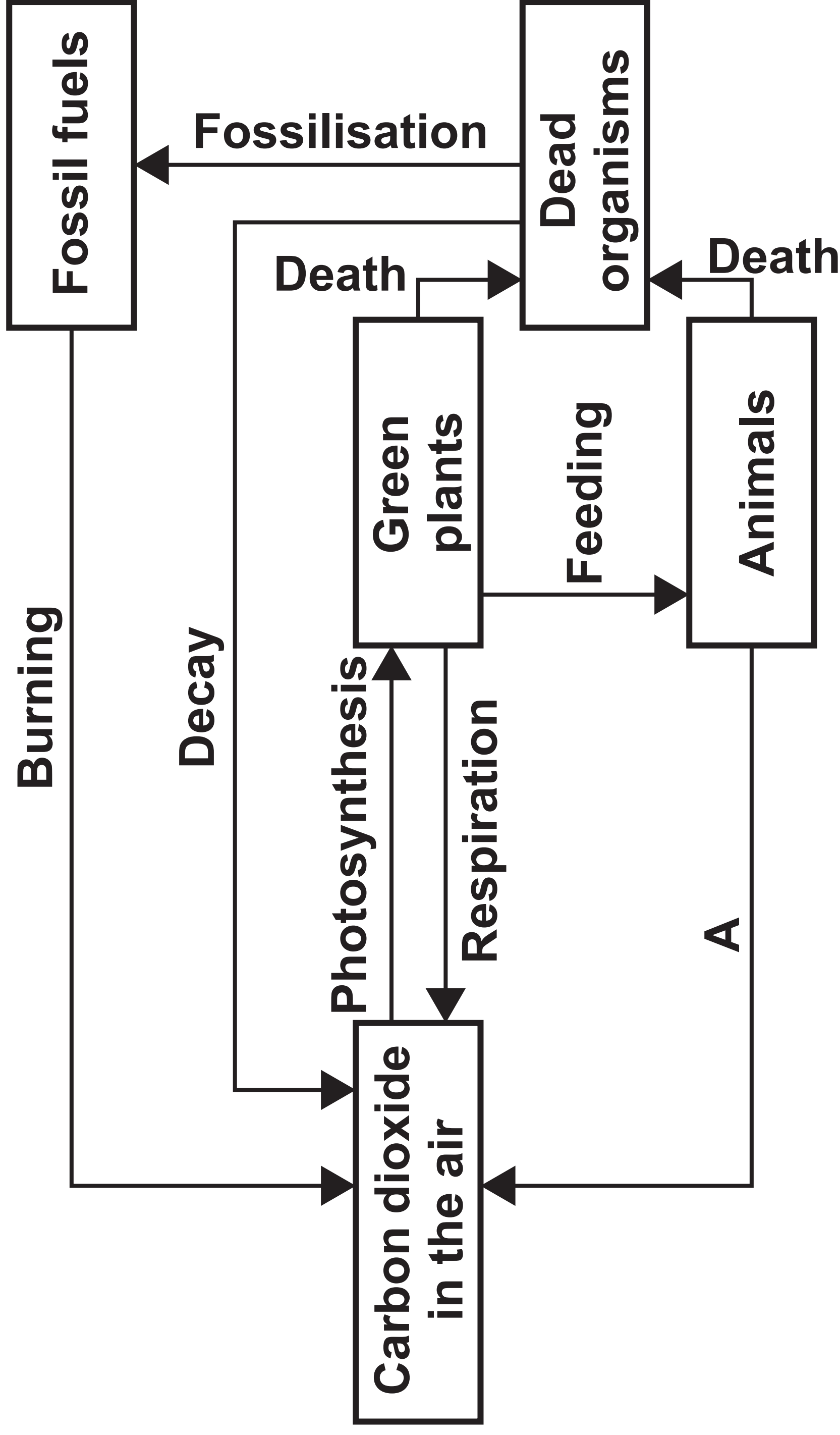


TABLE 2.2

Distance travelled (km)	20
Mass of carbon dioxide produced per person travelling by car (kg)	1.20
Mass of carbon dioxide produced per person travelling by bus (kg)	0.20

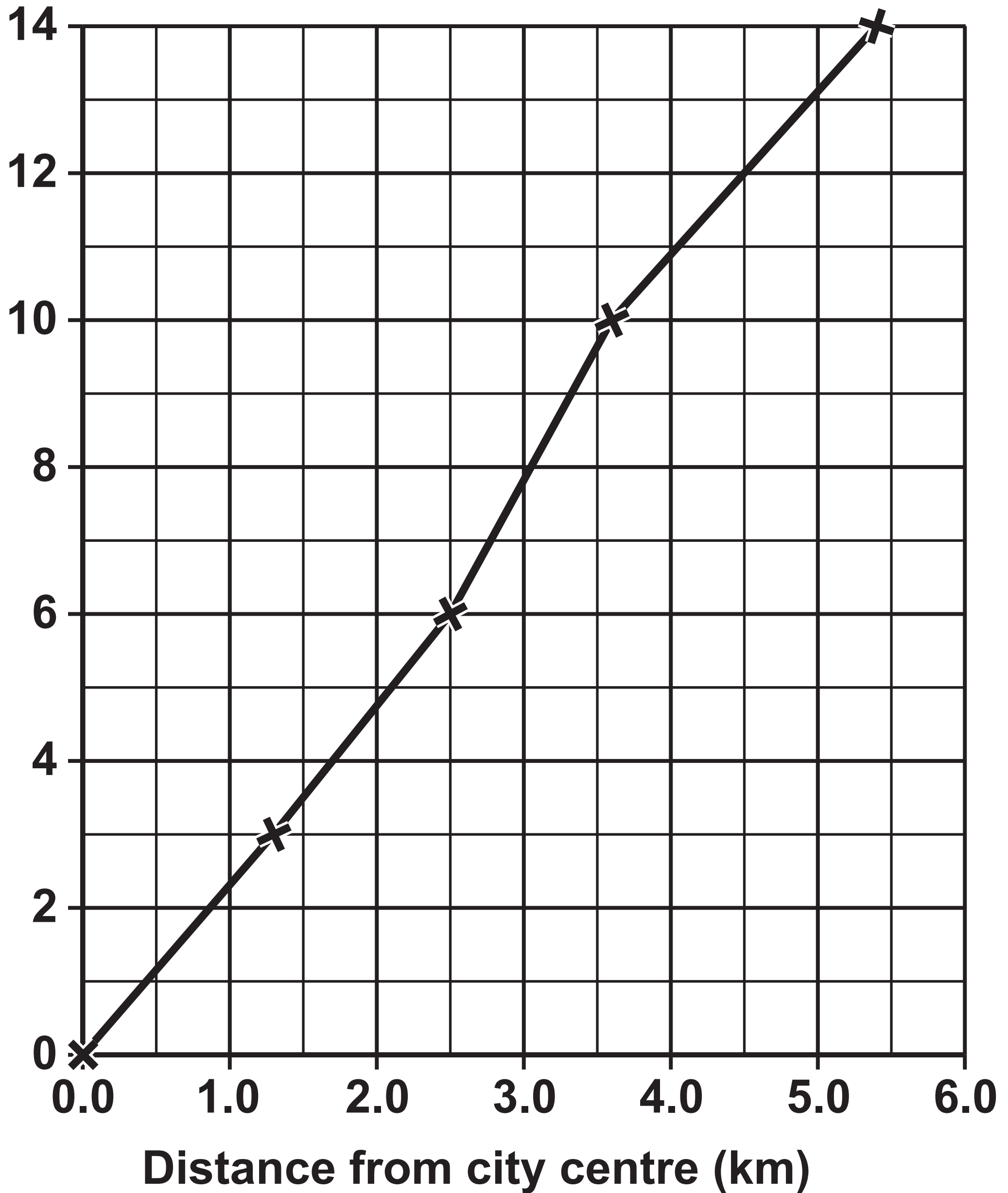
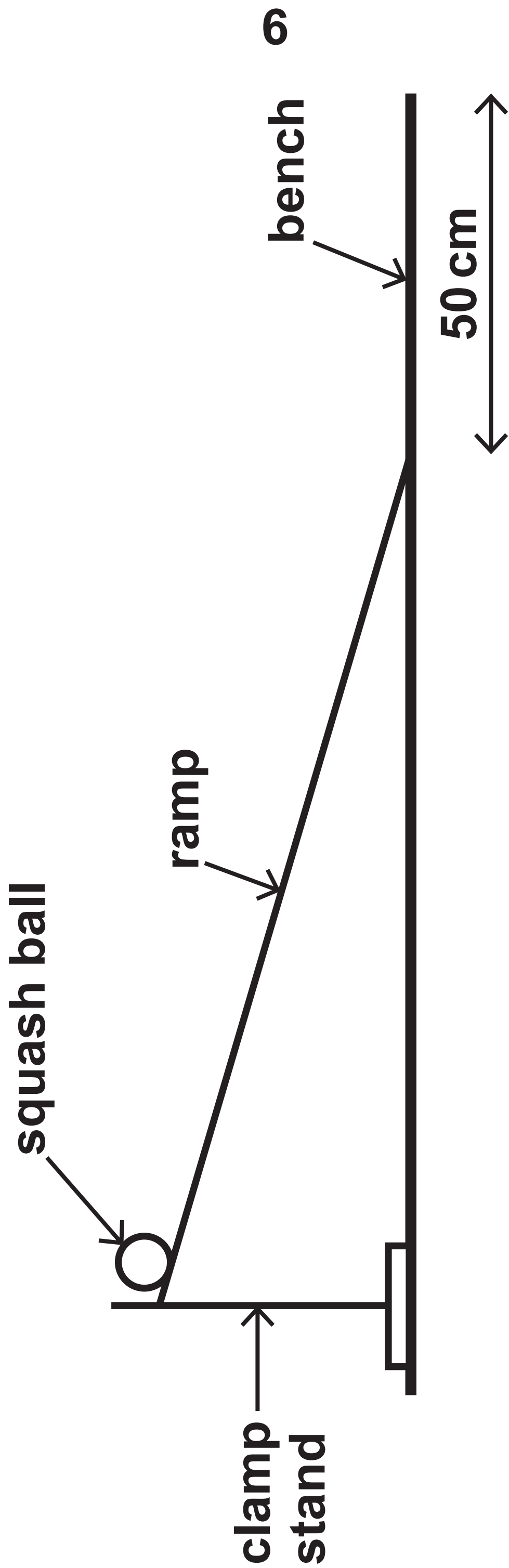
GRAPH 3.1**Abundance of lichen (units)**

DIAGRAM 4.1



PUNNETT SQUARE 5.1

	F	f

PUNNETT SQUARE 5.2

	F	f
F	FF	Ff
f	Ff	ff

TABLE 5.3

Time (minutes)	0	25	30	45	60
Number of bacteria	1	2		8	

DIAGRAM 6.1

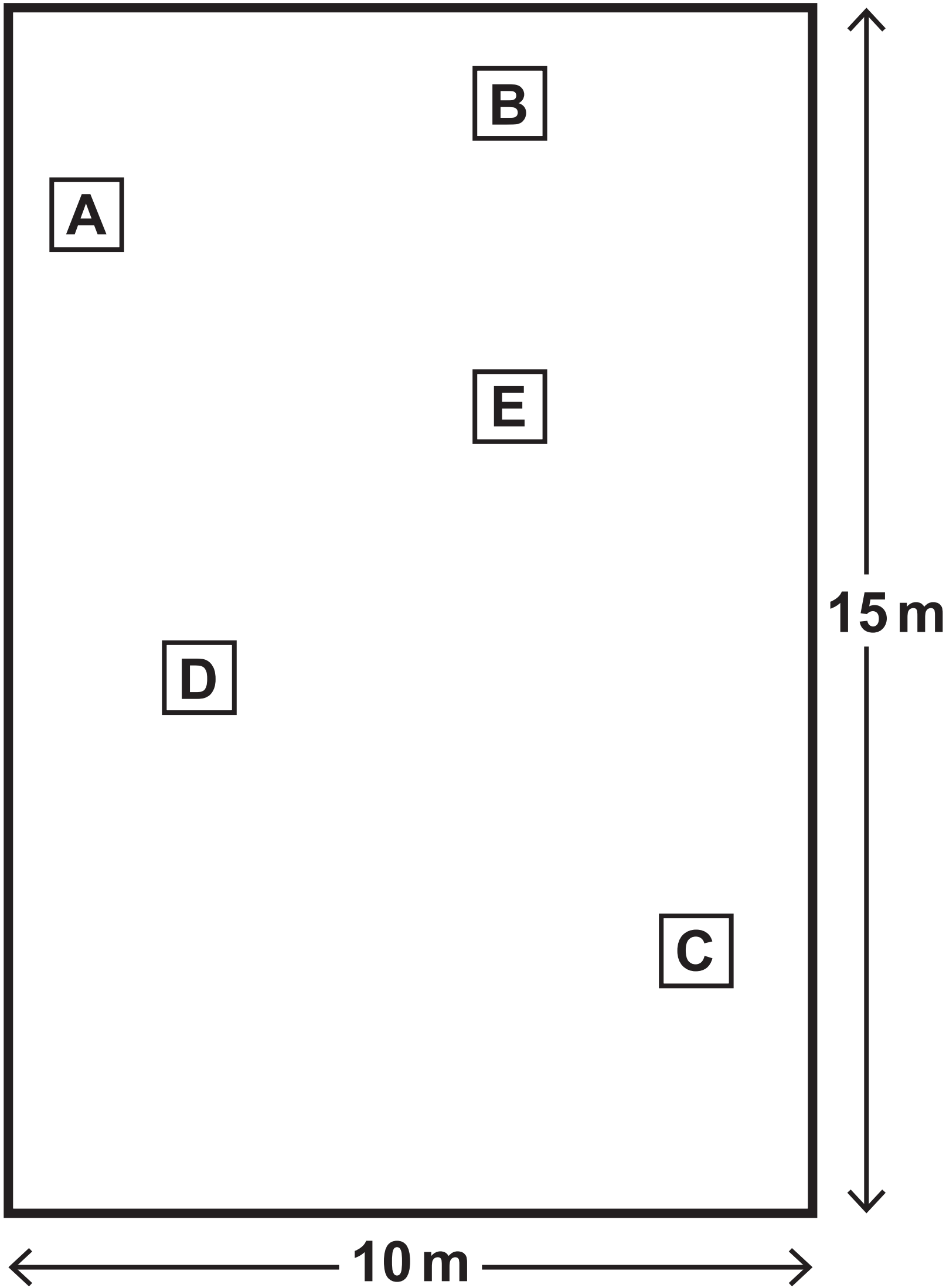


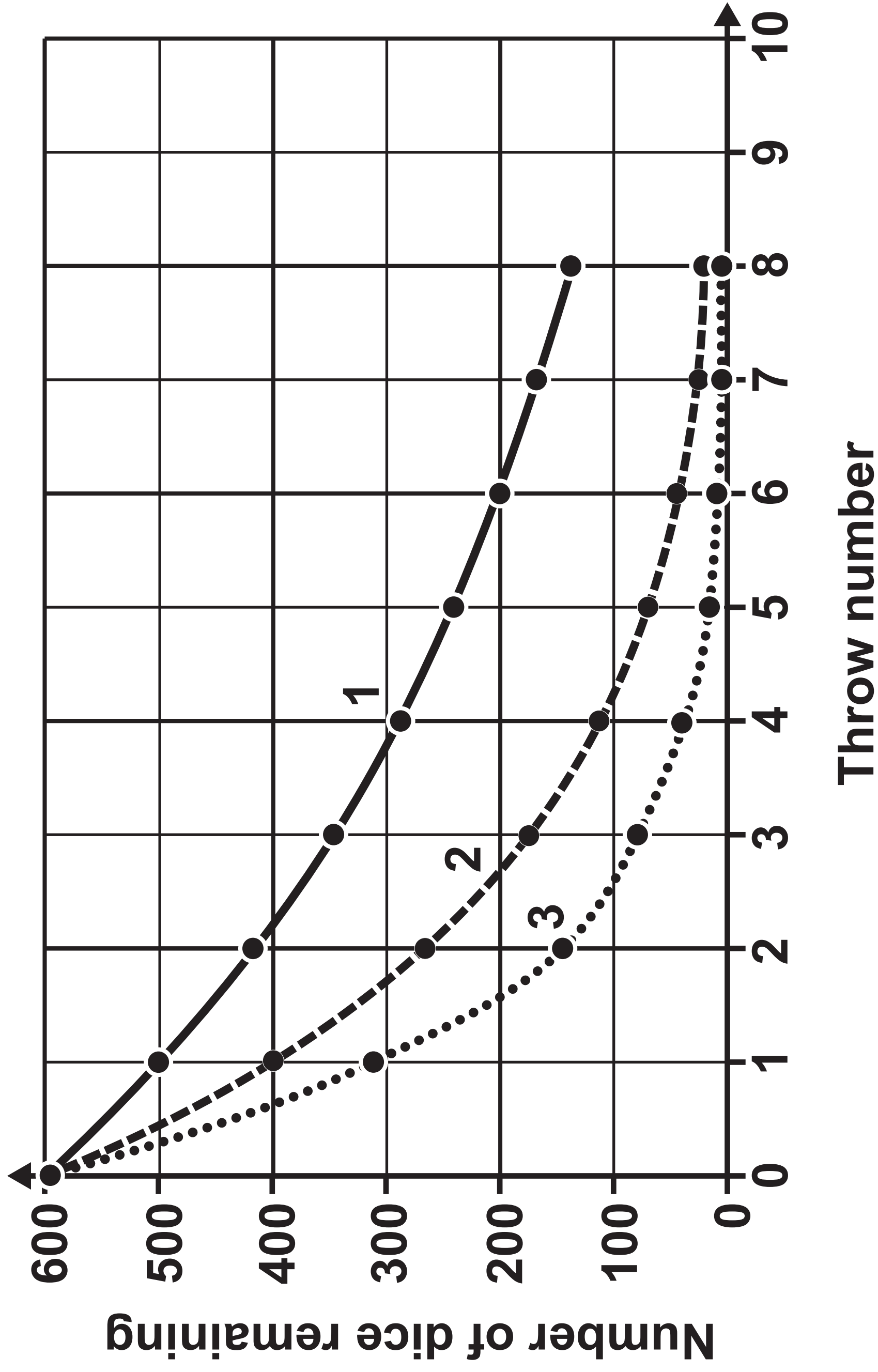
TABLE 6.2

Quadrat	Number of dandelions
A	6
B	3
C	4
D	8
E	4

DIAGRAM 8.1



GRAPH 9.1





3445U20-1

TUESDAY, 13 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life

FOUNDATION TIER

Data Booklet

THE PERIODIC TABLE

1

2

GROUP

1
H
1

7 Li 3	9 Be 4
23 Na 11	24 Mg 12
39 K 19	40 Ca 20
86 Rb 37	88 Sr 38
133 Cs 55	137 Ba 56
223 Fr 87	226 Ra 88

KEY

Ar

Sym

Z

relative atomic mass

symbol

atomic number

45 Sc 21	48 Ti 22	51 V 23	52 Cr 24	55 Mn 25	56 Fe 26	59 Co 27
89 Y 39	91 Zr 40	93 Nb 41	96 Mo 42	99 Tc 43	101 Ru 44	103 Rh 45
139 La 57	179 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77
227 Ac 89						

3

4

5

6

7

0

$${}^4_2\text{He}$$

**11
B
5**

12
C
6

14
N
7

16
0
8

19
F
9

20
Ne
10

27
Al
13

28
Si
14

**31
P
15**

32
S
16

35.5
Cl
17

40
Ar
18

59
Ni
28

63.5
Cu
29

65
Zn
30

70
Ga
31

73
Ge
32

75
As
33

79
Se
34

80
Br
35

84
Kr
36

106
Pd
46

108
Ag
47

112
Cd
48

115
In
49

119
Sn
50

122
Sb
51

128
Te
52

127
I
53

131
Xe
54

195
Pt
78

**197
Au
79**

201
Hg
80

**204
TL
81**

207
Pb
82

209
Bi
83

210
Po
84

210
At
85

222
Rn
86

THE PERIODIC TABLE

PERIODIC TABLE – KEY

ATOMIC NUMBER – SYMBOL – NAME

1	H – Hydrogen	19	K – Potassium
2	He – Helium	20	Ca – Calcium
3	Li – Lithium	21	Sc – Scandium
4	Be – Beryllium	22	Ti – Titanium
5	B – Boron	23	V – Vanadium
6	C – Carbon	24	Cr – Chromium
7	N – Nitrogen	25	Mn – Manganese
8	O – Oxygen	26	Fe – Iron
9	F – Fluorine	27	Co – Cobalt
10	Ne – Neon	28	Ni – Nickel
11	Na – Sodium	29	Cu – Copper
12	Mg – Magnesium	30	Zn – Zinc
13	Al – Aluminium	31	Ga – Gallium
14	Si – Silicon	32	Ge – Germanium
15	P – Phosphorus	33	As – Arsenic
16	S – Sulfur	34	Se – Selenium
17	Cl – Chlorine	35	Br – Bromine
18	Ar – Argon	36	Kr – Krypton

37	Rb – Rubidium	57	La – Lanthanum
38	Sr – Strontium	72	Hf – Hafnium
39	Y – Yttrium	73	Ta – Tantalum
40	Zr – Zirconium	74	W – Tungsten
41	Nb – Niobium	75	Re – Rhenium
42	Mo – Molybdenum	76	Os – Osmium
43	Tc – Technetium	77	Ir – Iridium
44	Ru – Ruthenium	78	Pt – Platinum
45	Rh – Rhodium	79	Au – Gold
46	Pd – Palladium	80	Hg – Mercury
47	Ag – Silver	81	Tl – Thallium
48	Cd – Cadmium	82	Pb – Lead
49	In – Indium	83	Bi – Bismuth
50	Sn – Tin	84	Po – Polonium
51	Sb – Antimony	85	At – Astatine
52	Te – Tellurium	86	Rn – Radon
53	I – Iodine	87	Fr – Francium
54	Xe – Xenon	88	Ra – Radium
55	Cs – Caesium	89	Ac – Actinium
56	Ba – Barium		